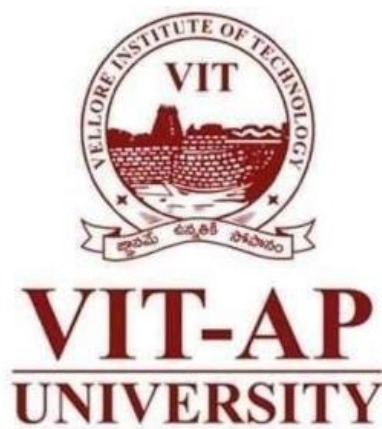


SWE4004 - Cloud Computing and its Applications

Topic: Project Report



Submitted to

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(Integrated M.Tech in Software Engineering)

Course Platform: Google Cloud Skills Boost

Course1: Google Cloud Computing Foundations - Cloud Computing Fundamentals

Course hours: 8 hours

Course Type: Completion Badge

Introduction:

The course "**Google Cloud Computing Foundations: Cloud Computing Fundamentals**" is an entry-level course offered through the **Google Skill Boost** platform. It serves as the first step into the world of cloud computing, helping learners understand the basic concepts, terminology, and services that power modern cloud infrastructures.

This course introduces the core benefits of cloud computing, including scalability, flexibility, and cost-efficiency. It provides insights into various service models such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS), and explains how these models are used by businesses today. The course also explores essential components like data centres, virtual machines, cloud storage, and networking fundamentals, forming a solid theoretical base before diving into more advanced Google Cloud topics.

By the end of the course, learners develop a clear understanding of how cloud computing has transformed the IT industry and gain the vocabulary and foundational knowledge needed to explore practical cloud services in later modules or certifications. I chose this course because cloud computing is a critical skill in today's tech landscape, and Google Cloud is one of the leading cloud service providers alongside AWS and Azure. As a student pursuing an integrated M.Tech degree, I believe that understanding the fundamentals of cloud infrastructure is essential for both my academic growth and future career in software engineering or data systems.

Additionally, I wanted to:

- Build a strong foundation before moving into hands-on labs and advanced cloud topics.
- Understand how businesses use the cloud to manage resources efficiently.
- Prepare for future certifications like Associate Cloud Engineer or Cloud Digital Leader.
- Align my learning with current industry trends such as remote computing, cloud-native development, and scalable systems.

Choosing this course was a strategic decision to strengthen my core understanding of IT infrastructure, while also positioning myself to confidently explore real-world applications in cloud environments.

Module-Wise Analysis of the Course Content:

No.of Modules: 3

Module1

So, What is the cloud anyway?

Objective:

In the simplest terms, the cloud refers to computing services and resources delivered over the internet. Instead of storing data or running programs on your personal computer or local servers, the cloud allows you to access these resources remotely through data centers managed by cloud providers like Google Cloud, AWS, or Microsoft Azure.

Cloud computing includes a wide range of services like storage, databases, servers, networking, software, and analytics. These services are delivered on-demand and typically follow a pay-as-you-go model, meaning users only pay for what they use.

The cloud has become a foundation for modern digital transformation. It enables companies to scale their operations, reduce IT costs, improve accessibility, and innovate faster without needing to invest in expensive physical infrastructure.

Lesson1: Cloud Computing

Objective

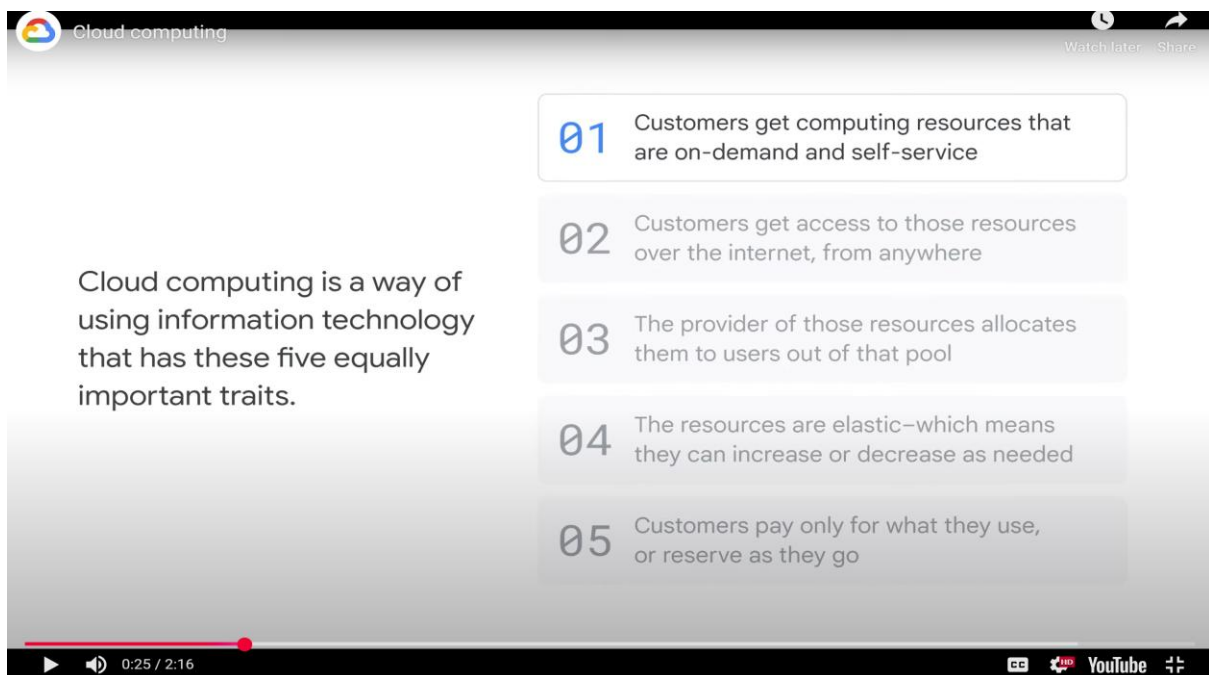
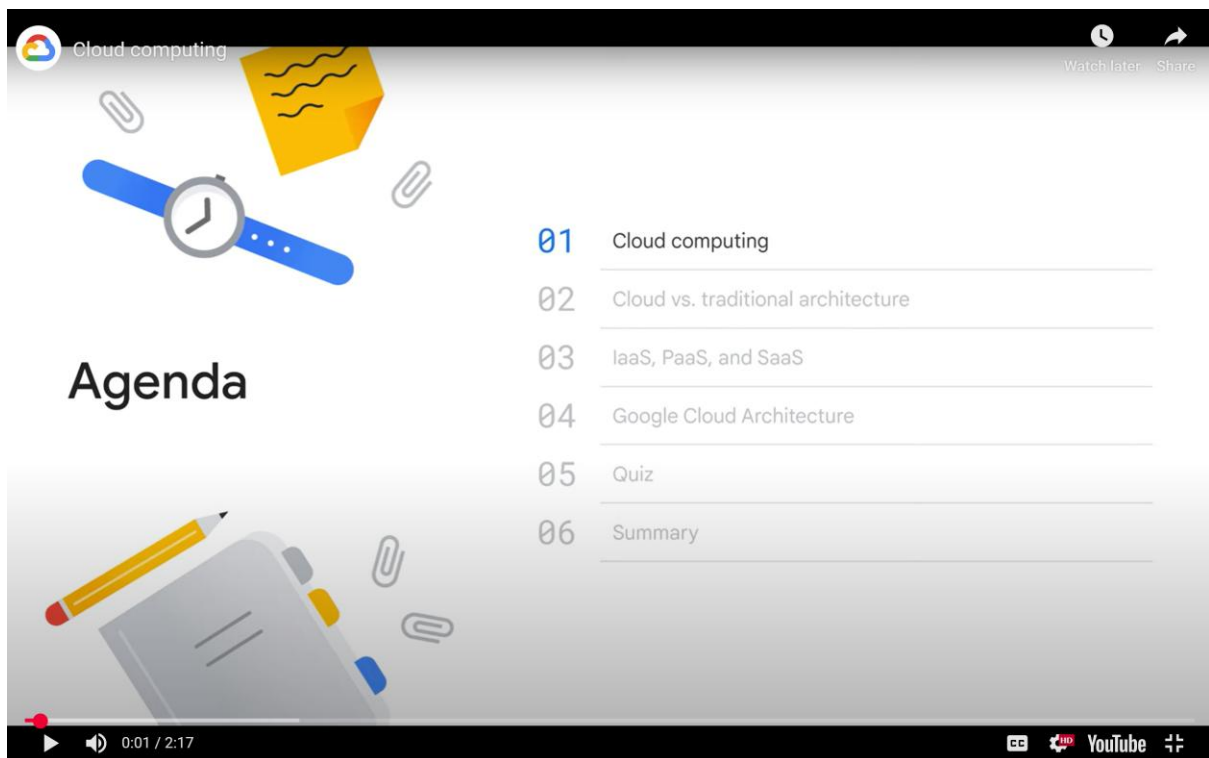
To understand the core idea of cloud computing, its definition, and why businesses are rapidly adopting it.

Key Concepts

- Cloud computing delivers computing services like storage, servers, and applications over the internet.
- It replaces the need for physical infrastructure with scalable and on-demand resources.
- Users can access data and apps anytime, from anywhere, using just an internet connection.

My Learning

I learned that cloud computing is not just about storing files online; it's about transforming how businesses and individuals access and use technology. It replaces traditional IT systems with a more agile, scalable, and cost-efficient alternative. This also made me realize how cloud platforms have fundamentally changed how companies operate, providing the flexibility to scale and innovate quickly. I now understand the key service models (IaaS, PaaS, SaaS) and how they each cater to different business needs.



Lesson2: Cloud vs Traditional Architecture

Objective

To compare cloud computing with traditional IT infrastructure and understand the advantages of the cloud.

Key Concepts

- **Traditional IT Architecture:** Involves physical servers, storage, and software that are managed on premise by an IT team. Scaling requires physical upgrades.
- **Cloud Architecture:** Utilizes virtualized resources over the internet, hosted by cloud providers. Scaling, updates, and maintenance are handled by them.
- **Cost Efficiency:** Traditional IT requires high upfront costs for hardware. Cloud computing offers a pay-as-you-go model, lowering initial investment.
- **Scalability:** The cloud scales easily based on demand, while traditional IT requires manual additions of hardware and software.
- **Maintenance:** In traditional IT, maintenance is done in-house, whereas in the cloud, maintenance and updates are automatic and managed by the provider.
- **Disaster Recovery:** Cloud platforms offer built-in redundancy and automatic disaster recovery, while traditional IT needs separate disaster recovery setups.

My Learning

I learned that the cloud offers more flexibility and cost-efficiency than traditional IT systems, which require higher upfront costs and manual management. The cloud allows for easy scaling and automatic updates, making it a more efficient solution.

The image shows a YouTube video player interface. The video title is "Cloud vs. traditional architecture". The video player has a play button, a progress bar at 0:02 / 3:11, and a volume icon. The video content displays an "Agenda" list with six items: 01 Cloud computing, 02 Cloud vs. traditional architecture (highlighted in blue), 03 IaaS, PaaS, and SaaS, 04 Google Cloud Architecture, 05 Quiz, and 06 Summary. The background of the video content features a light gray background with illustrations of a blue wristwatch, a yellow notepad, a yellow pencil, and paper clips.

Cloud vs. traditional architecture

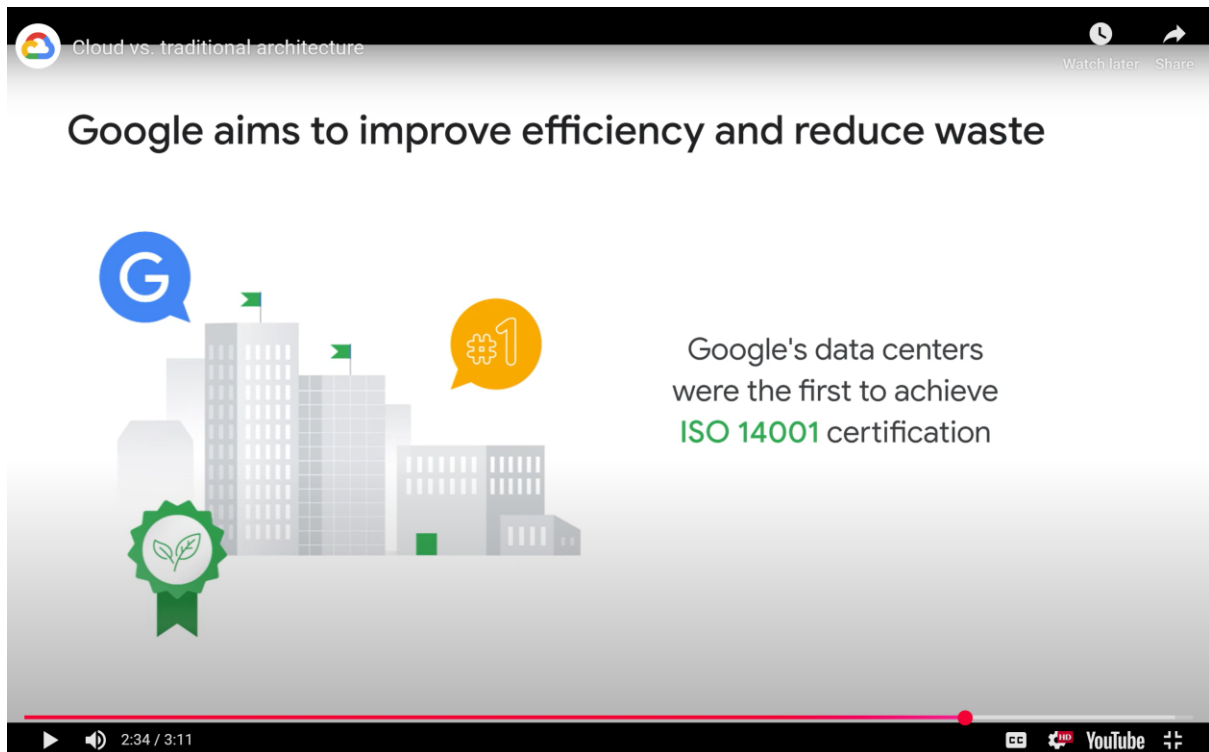
Watch later Share

Agenda

- 01 Cloud computing
- 02 Cloud vs. traditional architecture
- 03 IaaS, PaaS, and SaaS
- 04 Google Cloud Architecture
- 05 Quiz
- 06 Summary

0:02 / 3:11

CC BY YouTube



Lesson 3: IaaS, PaaS, SaaS

Objective

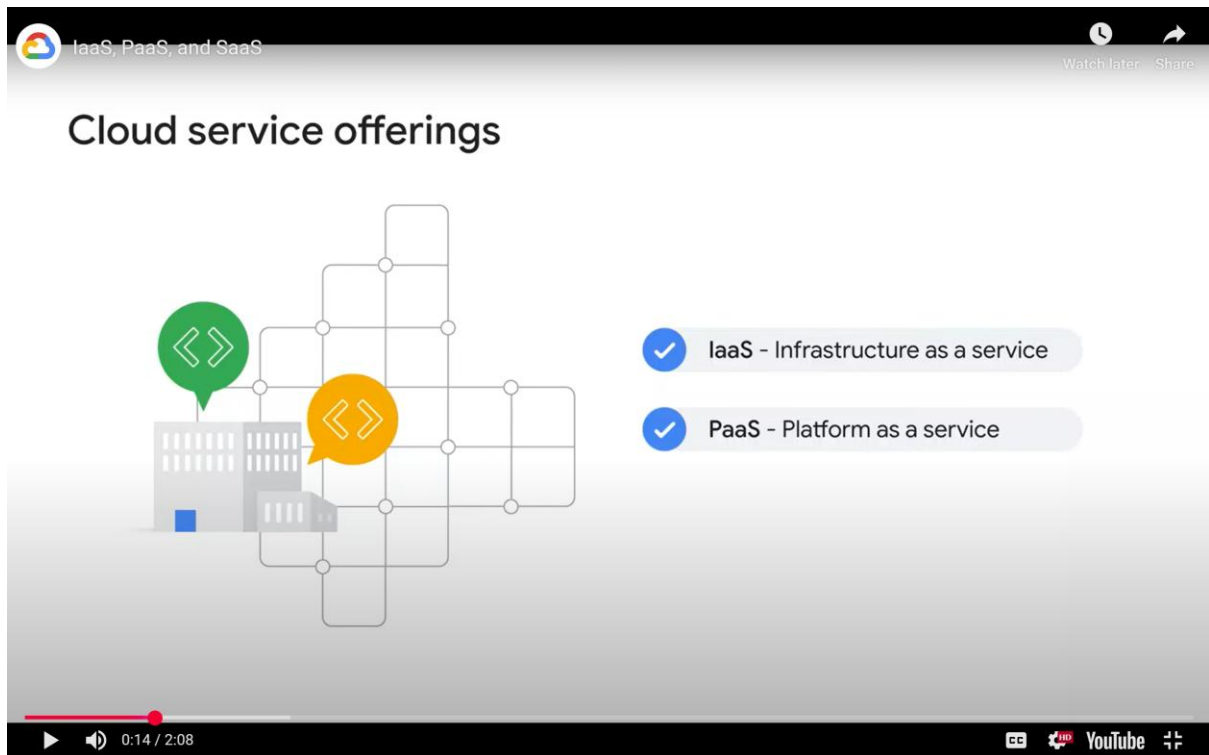
To understand the three main cloud service models: IaaS, PaaS, and SaaS.

Key Concepts

- **IaaS (Infrastructure as a Service):** Provides virtualized computing resources like servers, storage, and networking. Users manage the OS and applications. Example: Google Compute Engine.
- **PaaS (Platform as a Service):** Offers a platform for building, testing, and deploying applications without managing the underlying infrastructure. Example: Google App Engine.
- **SaaS (Software as a Service):** Delivers software applications over the internet, where users access them via a browser. No installation or maintenance is required. Example: Gmail.

My Learning

I learned that each model provides different levels of control and management. IaaS gives more control over infrastructure, PaaS focuses on application development, and SaaS delivers ready-to-use software.



Lesson4: Google Cloud Architecture

Objective

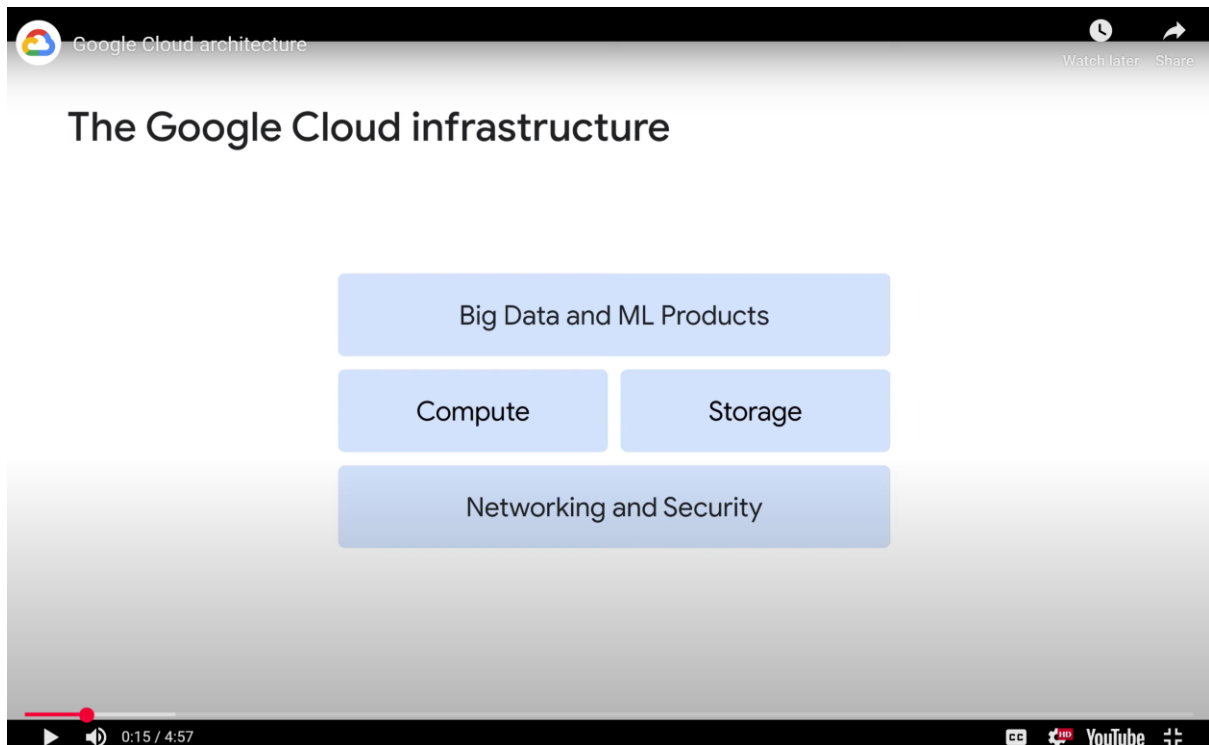
To understand the basic structure and components of Google Cloud Platform (GCP).

Key Concepts

- **Regions and Zones:** GCP has data centers around the world. A region is a specific geographical area, and each region has multiple zones for high availability.
- **Projects and Resources:** Everything in GCP is organized under projects. Resources like virtual machines, storage, and databases are created and managed within these projects.
- **Global Network:** GCP uses Google's global private fibre network, offering high speed and low latency.
- **IAM (Identity and Access Management):** Helps control who can access which resources, ensuring secure cloud operations.

My Learning

I learned how GCP is structured globally for performance and reliability. The use of regions, zones, and IAM makes it secure, scalable, and suitable for businesses of all sizes.



Lesson4:Quiz

Google Cloud Skills Boost

← Main menu

Google Cloud Computing Foundations: Cloud Computing Fundamentals

Course - 8 ✓ Complete hours

So, what is the cloud anyway? ^

- ✓ Introduction
- ✓ Cloud computing
- ✓ Cloud vs. traditional architecture
- ✓ IaaS, PaaS, and SaaS
- ✓ Google Cloud architecture
- ✓ **Quiz**

Google Cloud Computing Foundations: Cloud Computing Fundamentals > So, what is the cloud anyway?

Quiz

Your score: 100% Passing score: 80% [Retake](#)

Congratulations! You passed this assessment.

✓ 1. What is the fully automated, elastic third-wave cloud that consists of a combination of automated services and scalable data?

- ☐ Colocation
- ✓ ☒ Container-based architecture
- ☐ On-premises
- ☐ Virtualized data center

Good job! That's the correct answer.

[← Previous](#) [Next >](#)

Module2

Start with a Solid Platform

Objective:

To explore the core services of Google Cloud used to build a strong, scalable, and secure infrastructure. Of course! Here's the same paragraph without any bold formatting

The module "Start with a Solid Platform" focuses on helping learners build a strong foundation in using Google Cloud tools and interfaces. It begins with an introduction to the Google Cloud Console, a web-based interface to manage services. The lesson on understanding projects explains how resources are grouped and managed in GCP using projects. In Google Cloud Billing, the concepts of account setup, budgeting, and cost tracking are discussed. The module also includes lessons on how to install and configure the Google Cloud SDK, enabling the use of GCP services from the command line. Cloud Shell and Getting Started with Cloud Shell and gcloud provide practical insights into using GCP's browser-based command-line tool for development and management tasks. A Tour of Google Cloud Hands-on Lab offers an interactive experience to explore the platform. Google Cloud APIs teaches how APIs can be enabled and used in GCP, and the Cloud Console Mobile App introduces the mobile interface for on-the-go resource monitoring and management. The module concludes with a review quiz to assess understanding. Additionally, two hands-on labs are included: one focusing on using the Cloud Console and the other on creating and managing a project using Cloud Shell. These labs provide practical exposure to navigating and working with core cloud tools.

Lab1: A Tour of Google Cloud Hands-on Lab

Duration Time: 45 minutes

Objective:

The objective of this lab is to provide a guided, interactive introduction to the Google Cloud Platform. It helps users explore the Cloud Console, understand the layout, and perform basic tasks. The lab is designed to build familiarity with key tools and navigation.

Steps:

Task 1: Access the Cloud console

Start the lab

It may take a moment for the Google Cloud environment and credentials to spin up. When the timer starts counting down and the Start Lab button changes to a red End Lab button, everything is in place and you're ready to sign in to the Cloud console.

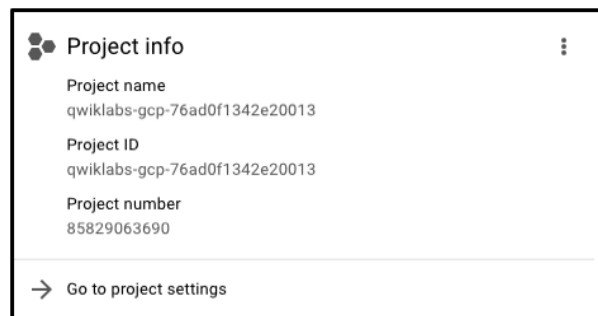
Sign in to Google Cloud

Now that you have a better understanding of the Lab details pane, use its contents to sign in to the Cloud Console.

- Click Open Google Cloud console.
- This will open the Google Cloud sign-in page in a new browser tab.
- If you have used a Google application like Gmail before, the sign-in page will look familiar.
- Tip: Open the tabs in separate windows, side-by-side, to make it easier to follow the instructions while working in the Cloud Console.
- The Username from the Lab Details pane automatically fills in. Click Next.
- Copy the Password from the Lab details pane, paste it in the Password field, and click Next.
- Click I understand to indicate your acknowledgement of Google's terms of service and privacy policy.
- On the Welcome page, check Terms of Service to agree to Google Cloud's terms of service, and click Agree and continue.

Task 2: View Projects in the cloud console

The upper-left corner of the central pane contains a card labelled Project info.



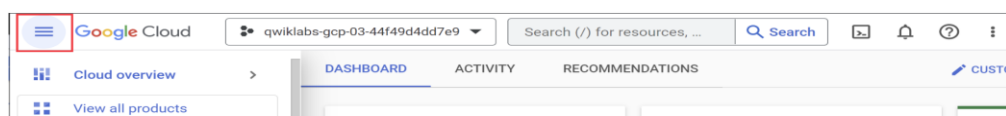
View all projects

In this lab, you actually have access to more than one Google Cloud project. In fact, in some labs, you may be given more than one project in order to complete the assigned tasks.

- In the Google Cloud console title bar, next to your project name, click the drop-down menu.
- In the Select a project dialog, click All.
- The resulting list of projects includes a "Qwiklabs Resources" project.
- Click Cancel to return to the Cloud overview page.

Navigation menu and services

The Google Cloud console title bar also contains the Navigation menu.



Clicking this icon opens (or hides) the Navigation menu that provides quick access to Google Cloud's core services.

- If the menu isn't open, click the Navigation menu.
- Click View all Products, then scroll through the categories of tools and services.

Task 3: Review and modify roles and permissions

View your roles and permissions

- On the Navigation menu, click IAM & Admin > IAM. This opens a page that contains a list of users, permissions, and roles granted to specific accounts.
- Find the student "@qwiklabs" username you signed in to the lab with.

Filter Enter property name or value							?	III
☐ Type	Principal ↑	Name	Role	Security insights ?	Inheritance			
☐	admiral@qwiklabs-services-prod.iam.gserviceaccount.com		Owner					
☐	qwiklabs-gcp-01-3ff16246f073@qwiklabs-gcp-01-3ff16246f073.iam.gserviceaccount.com	Qwiklabs User Service Account	BigQuery Admin Owner					
			Storage Admin					
☐	student-03-5d434f03c215@qwiklabs.net	student d7c20ab9	Editor Viewer					

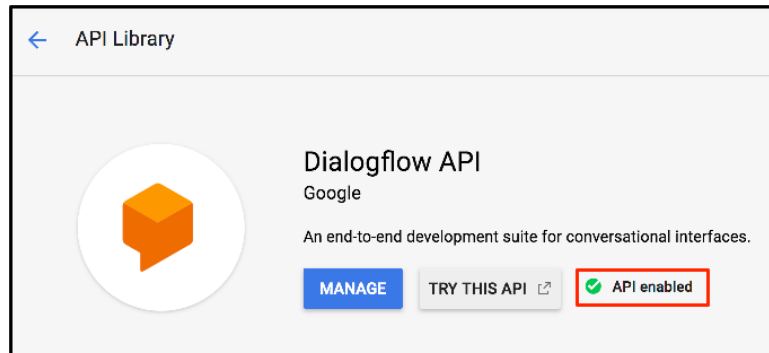
Grant an IAM role

- On the Navigation menu, click IAM & Admin > IAM.
- Click Grant access.
- In the Add principals section, enter an identifier User 2 for the principal.
- From the Select a role drop-down menu in the Assign roles section, search for Viewer, then click Viewer.
- Click Save.
- Verify that the principal and the corresponding role are listed on the IAM page.

You have successfully granted an IAM role to a second student account.

Task 4. Enable APIs and services

- On the Google Cloud Console, navigate to the Navigation menu and select APIs & Services > Dashboard.
- Click Enable APIs and Services at the top of the page.
- In the API Library, search for the desired API (e.g., Cloud Storage, Compute Engine) using the search bar.
- Select the API from the search results, then click Enable to activate the service for your project.



Task5: End Your Lab

Congratulations!

In just 30 minutes, you developed a solid understanding of the Cloud console and the platform's key features. You learned about projects, roles, and the types of services the platform offers. You also practiced with Cloud IAM and the API libraries. You are now ready to take more labs.

Lab2: Getting started with cloudshell and gcloud

Duration Time: 45 minutes

Objective:

The objective of the Getting Started with Cloud Shell and gcloud lab is to introduce Cloud Shell and teach basic usage of the gcloud command-line tool for managing Google Cloud resources.

Steps:

Task1: Configuring your environment

- Set the project ID using:
`gcloud config set project PROJECT_ID`
- Set the compute region:
`gcloud config set compute/region REGION`
- Set the compute zone:
`gcloud config set compute/zone ZONE`
- View the current project configuration:
`gcloud config get-value project`
`gcloud config get-value compute/region`
`gcloud config get-value compute/zone`
- Export environment variables:
`export PROJECT_ID=$(gcloud config get-value project)`
`export ZONE=$(gcloud config get-value compute/zone)`
- Verify environment variables:
`echo -e "PROJECT ID: $PROJECT_ID\nZONE: $ZONE"`

- Enable required APIs using the Cloud Console or gcloud commands.

```
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-123456789012/zones/us-central1-a/instances/gcelab2]
NAME      ZONE      MACHINE_TYPE  PREEMPTIBLE  INTERNAL_IP  EXTERNAL_IP
gcelab2   ZONE      e2-medium     10.128.0.2    34.67.152.90
```

Task 2: Filtering command-line output using gcloud:

- List all VM instances:
gcloud compute instances list
- Filter VM instances by zone:
gcloud compute instances list --filter="zone:(us-central1-a)"
- Display only the names of the VM instances:
gcloud compute instances list --format="value(name)"
- Get a table of VM names and their internal IPs:
gcloud compute instances list --format="table(name, networkInterfaces[0].networkIP)"

```
NAME: gcelab2
ZONE: ZONE
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.142.0.2
EXTERNAL_IP: 35.237.43.111
STATUS: RUNNING
```

NAME	NETWORK	DIRECTION	PRIORITY	ALLOW
default-allow-icmp	default	INGRESS	65534	icmp
default-allow-internal	default	INGRESS	65534	tcp:0-
default-allow-rdp	default	INGRESS	65534	tcp:33-
default-allow-ssh	default	INGRESS	65534	tcp:22-
dev-net-allow-ssh	dev-network	INGRESS	1000	tcp:22-
serverless-to-vpc-connector	dev-network	INGRESS	1000	icmp,
vpc-connector-egress	dev-network	INGRESS	1000	icmp,
vpc-connector-health-check	dev-network	INGRESS	1000	tcp:60-
vpc-connector-to-serverless	dev-network	EGRESS	1000	icmp,

NAME	NETWORK	DIRECTION	PRIORITY	ALLOW
default-allow-icmp	default	INGRESS	65534	icmp
default-allow-internal	default	INGRESS	65534	tcp:0-
default-allow-rdp	default	INGRESS	65534	tcp:33-
default-allow-ssh	default	INGRESS	65534	tcp:22-

Task 3: Connecting to your VM instance:

- Use the following command to connect to your VM using SSH:
`gcloud compute ssh gcelab2 --zone=$ZONE`
- Wait for the connection to establish. You may need to authorize the SSH key creation.
- Once connected, you will have terminal access to your virtual machine.
- You can now run commands directly on the VM to configure or manage it as needed.

Task 4: Updating the firewall:

- Create a firewall rule to allow HTTP traffic:
`gcloud compute firewall-rules create allow-http --allow tcp:80 --source-ranges=0.0.0.0/0 --target-tags=http-server`
- Tag your VM to apply the firewall rule:
`gcloud compute instances add-tags gcelab2 --tags=http-server --zone=$ZONE`
- Verify the firewall rule is applied correctly by checking the instance details in the Cloud Console or running:
`gcloud compute instances describe gcelab2 --zone=$ZONE`
- Now, the VM can receive HTTP traffic on port 80 from any source.

NAME	NETWORK	DIRECTION	PRIORITY	ALLOW	DENY
DISABLED					
default-allow-http	default	INGRESS	1000	tcp:80	
False					

Task 5: Viewing the system logs steps:

1. Connect to your VM:
`gcloud compute ssh gcelab2 --zone=$ZONE`
2. View system logs:
`sudo journalctl`
3. To filter by time:
`sudo journalctl --since "10 minutes ago"`
4. Exit the viewer with Ctrl+C.

```
NAME: projects/qwiklabs-gcp-01-4b75909db302/logs/GCEGuestAgent
NAME: projects/qwiklabs-gcp-01-4b75909db302/logs/OSConfigAgent
NAME: projects/qwiklabs-gcp-01-4b75909db302/logs/autoscaler.googleapis.com
NAME: projects/qwiklabs-gcp-01-4b75909db302/logs/cloudaudit.googleapis.com
NAME: projects/qwiklabs-gcp-01-4b75909db302/logs/cloudaudit.googleapis.com
NAME: projects/qwiklabs-gcp-01-4b75909db302/logs/cloudaudit.googleapis.com
NAME: projects/qwiklabs-gcp-01-4b75909db302/logs/compute.googleapis.com
NAME: projects/qwiklabs-gcp-01-4b75909db302/logs/compute.googleapis.com
NAME: projects/qwiklabs-gcp-01-4b75909db302/logs/compute.googleapis.com
NAME: projects/qwiklabs-gcp-01-4b75909db302/logs/run.googleapis.com%2
NAME: projects/qwiklabs-gcp-01-4b75909db302/logs/run.googleapis.com%2
```

Quiz

The screenshot shows a quiz interface for 'Google Cloud Computing Foundations: Cloud Computing Fundamentals'. The left sidebar lists course items, with 'Quiz' selected at the bottom. The main area displays the quiz title, score (80%), and a congratulatory message. A single question is shown with four options: Reports, Quotas (selected), Alerts, and Budgets. A green feedback box confirms the correct answer. Navigation buttons for 'Previous' and 'Next' are at the bottom.

Google Cloud Computing Foundations: Cloud Computing Fundamentals > Start with a solid platform

Quiz

Your score: 80% Passing score: 80% Retake

Congratulations! You passed this assessment.

✓ 1. Which billing tool is designed to prevent the over-consumption of resources due to an error or a malicious attack?

- ☐ Reports
- ☒ Quotas
- ☐ Alerts
- ☐ Budgets

Good job! That's the correct answer.

< Previous Next >

Module3

Use google cloud to build your apps

Objective:

The objective of this module is to introduce how to build, deploy, and manage applications using Google Cloud services. It focuses on using tools like App Engine and Cloud Functions for scalable app development. This module includes a total of 12 lessons, three hands-on labs, and one quiz. It guides learners through the process of building and deploying applications using Google Cloud services. The lessons cover essential tools such as App Engine, Cloud Functions, Cloud Storage, and Cloud Build, providing practical experience through labs to strengthen understanding.

Lab1: Cloud Run Functions: Qwik Start - Command Line

Duration Time: 30 minutes

Objective:

In the "Cloud Run Functions: Qwik Start - Command Line" lab, the objective is to help you understand how to deploy a simple HTTP-triggered Cloud Function using the Google Cloud command-line interface (gcloud CLI). You will write a basic function that responds to HTTP requests, deploy it using the CLI, and then test it by sending a request to the deployed endpoint. This lab provides hands-on experience with the fundamentals of Google Cloud Functions, including defining entry points, setting triggers, and using serverless architecture to run your code without managing infrastructure.

Steps:

Task 1: Create a Function

- In the Google Cloud Console, go to the Cloud Functions section.
- Click Create Function and choose the region.
- Set the trigger type (e.g., HTTP), and enter a name for your function.
- In the source code section, write or paste your function code.
- Click Deploy to create and launch your cloud function.

```
added 140 packages, and audited 141 packages in 9s

27 packages are looking for funding
run `npm fund` for details

found 0 vulnerabilities
```

Task 2: Deploy your function:

- Review the function details like name, region, trigger, and runtime.
- Click on Deploy to start the deployment process.
- Wait for the deployment to complete; a green checkmark indicates success.

```
BuildConfig:
  automaticUpdatePolicy: {}
  build: projects/630521560493/locations/REGION/builds/7ff9d415-50d9-
  dockerRegistry: ARTIFACT_REGISTRY
  dockerRepository: projects/PROJECT_ID/locations/REGION/repositories
  entryPoint: helloPubSub
  ...
State: ACTIVE
  ...
UpdateTime: '2024-08-05T13:51:05.317298824Z'
Url: https://REGION-PROJECT_ID.cloudfunctions.net/nodejs-pubsub-funct
```

Task 3: Test the function:

- After deployment, locate the function's trigger URL in the Google Cloud Console.
- Open a new browser tab or terminal and paste the trigger URL.
- If the function is an HTTP function, you should see the output or response in your browser or terminal.
- Verify that the function is working as expected based on the output.

```
messageIds:
- '11927162971409664'
```

Task 4: View logs

- In the Google Cloud Console, go to the Cloud Functions section.
- Find your deployed function and click on it.
- Under the Logs tab, you will see the execution logs for your function.
- You can filter logs by time, severity, or specific events to troubleshoot or verify function behavior.

```
LEVEL:
NAME: nodejs-pubsub-function
EXECUTION_ID: h4v6akxf4xt
TIME_UTC: 2024-08-05 15:15:25.723
LOG: Hello, Cloud Function Gen2!

LEVEL: I
NAME: nodejs-pubsub-function
EXECUTION_ID:
TIME_UTC: 2024-08-05 15:15:25.711
LOG:

LEVEL:
NAME: nodejs-pubsub-function
EXECUTION_ID: h4oxfjn7z1yu
TIME_UTC: 2024-08-05 15:16:34.383
LOG: Hello, Friend!

LEVEL: I
NAME: nodejs-pubsub-function
EXECUTION_ID:
TIME_UTC: 2024-08-05 15:16:34.291
LOG:
```

Congratulations!

You used the Google Cloud console to create, deploy, and test a Cloud Run function using the command line.

Lab2: App Engine- qwik start Python

Duration Time: 20 minutes

Objective:

The objective of the "App Engine: Qwik Start - Python" lab is to introduce you to deploying a basic Python web application on Google App Engine. In this lab, you will clone a simple "Hello, World!" app written in Python, set up your environment, and deploy the app using the gcloud command-line tool. The goal is to help you understand the fundamentals of App Engine, such as using app.yaml for configuration, deploying applications in a fully managed serverless environment, and viewing the deployed app through a browser. This hands-on experience serves as a starting point for building scalable web applications on Google Cloud.

Steps:

Task1: Enable Google App Engine Admin API

The App Engine Admin API enables developers to provision and manage their App Engine applications.

Steps to enable the API:

- In the left navigation menu, go to APIs & Services > Library.
- In the search box, type App Engine Admin API.
- Click on the App Engine Admin API card.
- Click Enable.
- If there is no prompt to enable the API, it means it is already enabled and no action is needed.

Task2: Download the Hello World App

There is a simple Hello World app for Python you can use to quickly get a feel for deploying an app to Google Cloud. Follow these steps to download Hello World to your Google Cloud instance.

Steps:

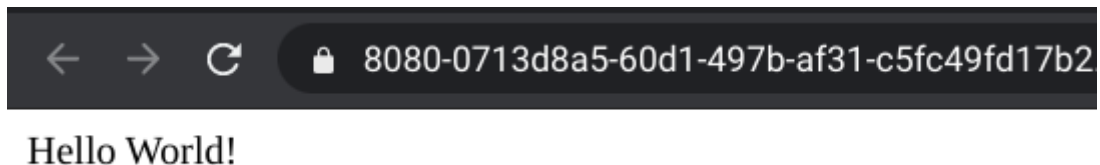
1. Enter the following command to copy the Hello World sample app repository to your Google Cloud instance:
`git clone https://github.com/GoogleCloudPlatform/python-docs-samples.git`
2. Go to the directory that contains the sample code:
`cd python-docs-samples/appengine/standard_python3/hello_world`
3. Set up the Python environment:
 - `sudo apt update`
 - `sudo apt install -y python3-venv`
 - `python3 -m venv myenv`
 - `source myenv/bin/activate`

Task3: Test the application

Test the application using the Google Cloud development server (`dev_appserver.py`), which is included with the preinstalled App Engine SDK.

Steps:

- From within your `hello_world` directory, where the app's `app.yaml` configuration file is located, start the Google Cloud development server with the following command:
`dev_appserver.py app.yaml`
- The development server is now running and listening for requests on port 8080.
- View the results by clicking the Web preview (web preview icon) > Preview on port 8080.
- You'll see this in a new browser window:



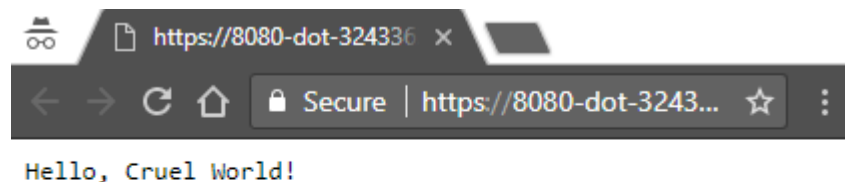
Task4: Manage a change

You can leave the development server running while you develop your application. The development server watches for changes in your source files and reloads them if necessary.

Let's try it. Leave the development server running. We'll open another command line window, then edit `main.py` to change "Hello World!" to "Hello, Cruel World!".

Steps:

1. Click the (+) next to your Cloud Shell tab to open a new command line session.
2. Enter this command to go to the directory that contains the sample code:
`cd python-docs-samples/appengine/standard_python3/hello_world`
3. Enter the following to open `main.py` in nano to edit the content:
`nano main.py`
4. Change "Hello World!" to "Hello, Cruel World!".
5. Save the file with CTRL-S and exit with CTRL-X.
6. Reload the Hello World! browser tab or click the Web Preview (web preview icon) > Preview on port 8080 to see the updated results.



Task5: Deploy your app

Steps:

- To deploy your app to App Engine, run the following command from within the root directory of your application where the `app.yaml` file is located:
`gcloud app deploy`
- Enter the number that represents your region: `<REGION>`
- The App Engine application will then be created.

```

Creating App Engine application in project [qwiklabs-gcp-
233dca09c0ab577b] and region ["REGION"]....done.
Services to deploy:

descriptor:      [/home/gcpstaging8134_student/python-docs-
samples/appengine/standard/hello_world/app.yaml]
source:          [/home/gcpstaging8134_student/python-docs-
samples/appengine/standard/hello_world]
target project:  [qwiklabs-gcp-233dca09c0ab577b]
target service:  [default]
target version:  [20171117t072143]
target url:      [https://qwiklabs-gcp-233dca09c0ab577b.appspot.com]

Do you want to continue (Y/n)?

```

- Enter Y when prompted to confirm the details and begin the deployment of service.

```

Beginning deployment of service [default]...
Some files were skipped. Pass `--verbosity=info` to see which ones.
You may also view the gcloud log file, found at
[/tmp/tmp.dYC7xGu3oZ/logs/2017.11.17/07.18.27.372768.log].

┌───────────────────────────────────────────────────────────────────────────────────┐
│ ≡ Uploading 5 files to Google Cloud Storage                                     ≡ │
└───────────────────────────────────────────────────────────────────────────────────┘
File upload
done.
Updating service [default]...done.
Waiting for operation [apps/qwiklabs-gcp-
233dca09c0ab577b/operations/2e88ab76-33dc-4aed-93c4-fdd944a95ccf] to
complete....done.
Updating service [default]...done.
Deployed service [default] to [https://qwiklabs-gcp-
233dca09c0ab577b.appspot.com]

You can stream logs from the command line by running:
$ gcloud app logs tail -s default

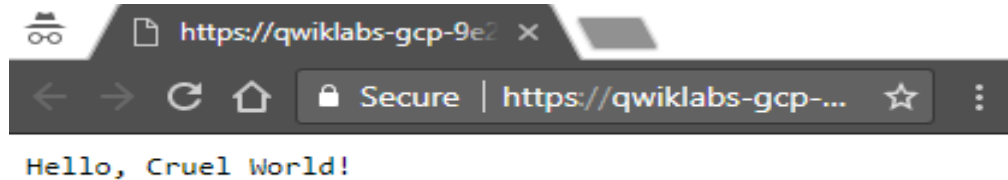
To view your application in the web browser run:
$ gcloud app browse

```

Task6: View your application

- To launch your browser enter the following command, then click on the link it provides:
 - gcloud app browse

Did not detect your browser. Go to this link to view your app:
<https://qwiklabs-gcp-233dca09c0ab577b.appspot.com>



- Your application is deployed and you can read the short message in your browser.

Congratulations!

Next steps /Learn more

- Learn more about an App Engine with [An Overview Of App Engine](#)
- Try something else with an App Engine with [Building a Python 3 Flask app on App](#)

Task 1. Create Google App Engine Admin API

Task 2. Download the Hello World app

Task 3. Test the application

Task 4. Make a change

Task 5. Deploy your app

Lab3: google kubernetes engine- Quick start

Duration Time: 45 minutes

Objective:

The objective of the Google Kubernetes Engine (GKE) - Quick Start lab is to introduce you to the basics of deploying and managing containerized applications using Kubernetes on Google Cloud. In this lab, you will create a Kubernetes cluster using GKE, deploy a sample containerized application to the cluster, and expose it to the internet using a LoadBalancer service. The goal is to help you understand core Kubernetes concepts such as pods, deployments, services, and how GKE simplifies container orchestration and scaling in a cloud environment.

Steps:

Task1: Set a default compute zone

Your compute zone is an approximate regional location in which your clusters and their resources live. For example, us-central1-a is a zone in the us-central1 region.

In your Cloud Shell session, run the following commands:

- Set the default compute region:
gcloud config set compute/region "REGION"
- Set the default compute zone:
gcloud config set compute/zone "ZONE"

```
Updated property [compute/region].
```

Task2: Create a GKE cluster

Steps:

- A cluster consists of at least one cluster master machine and multiple worker machines called nodes. Nodes are Compute Engine virtual machine (VM) instances that run the Kubernetes processes necessary to make them part of the cluster.
- Cluster names must start with a letter, end with an alphanumeric character, and cannot be longer than 40 characters.
- To create a cluster, run the following command:
- Create a cluster:
gcloud container clusters create --machine-type=e2-medium --zone=ZONE lab-cluster
- You can ignore any warnings in the output. It might take several minutes to finish creating the cluster.

```
NAME: lab-cluster
LOCATION: ZONE
MASTER_VERSION: 1.22.8-gke.202
MASTER_IP: 34.67.240.12
MACHINE_TYPE: e2-medium
NODE_VERSION: 1.22.8-gke.202
NUM_NODES: 3
STATUS: RUNNING
```

Task3: Get authentication credentials for the cluster

After creating your cluster, you need authentication credentials to interact with it.

- Authenticate with the cluster:
gcloud container clusters get-credentials lab-cluster

```
Fetching cluster endpoint and auth data.  
kubeconfig entry generated for my-cluster.
```

Task4: Deploy an application to the cluster

- You can now deploy a containerized application to the cluster. For this lab, you'll run hello-app in your cluster.
- GKE uses Kubernetes objects to create and manage your cluster's resources. Kubernetes provides the Deployment object for deploying stateless applications like web servers. Service objects define rules and load balancing for accessing your application from the internet.
- To create a new Deployment named hello-server from the hello-app container image, run the following command:
kubectrl create deployment hello-server --image=gcr.io/google-samples/hello-app:1.0

```
deployment.apps/hello-server created
```

- To create a Kubernetes Service, which is a Kubernetes resource that lets you expose your application to external traffic, run the following command:
kubectrl expose deployment hello-server --type=LoadBalancer --port 8080

In this command:

- --port specifies the port that the container exposes.
- type="LoadBalancer" creates a Compute Engine load balancer for your container.

```
service/hello-server exposed
```

- To inspect the hello-server Service, run kubectrl get: kubectrl get service

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP
hello-server	loadBalancer	10.39.244.36	35.202.234.26
kubernetes	ClusterIP	10.39.240.1	

- To view the application from your web browser, open a new tab and enter the following address, replacing [EXTERNAL IP] with the EXTERNAL-IP for hello-server.

Task5: Delete the cluster

- To delete the cluster, run the following command:
gcloud container clusters delete lab-cluster
- When prompted, type Y and press Enter to confirm.
- Deleting the cluster can take a few minutes. For more information, refer to the Google Kubernetes Engine (GKE) article on deleting a cluster.

Congratulations!

You have just deployed a containerized application to Google Kubernetes Engine! In this lab, you created a GKE cluster, deployed a containerized application to the cluster, created a Kubernetes service, and deleted the cluster. You can now apply this knowledge to deploy your own applications with GKE.

Quiz

The screenshot displays the Google Cloud Skills Boost interface. On the left is a navigation sidebar with a 'Main menu' link and a list of course topics including 'Start - Command Line', 'Cloud Run', 'Functions: Qwik Start - Command Line', 'Containerizing and Orchestrating Apps with GKE', 'Lab Intro: Kubernetes Engine: Qwik Start', 'Google Kubernetes Engine: Qwik Start', and 'Managed serverless computing with Cloud Run'. The 'Quiz' item is highlighted at the bottom of the sidebar. The main content area is titled 'Quiz' and shows 'Your score: 83%' and 'Passing score: 80%'. A 'Retake' button is visible. Below this, a message states 'Congratulations! You passed this assessment.' A single question is listed: '1. What is the Compute Engine feature that allows VMs to be added to or subtracted from an application based on load metrics?'. The options are 'Network time protocol (NTP)', 'Autoscaling' (which is selected with a green checkmark), 'Persistent disks', and 'Load balancing'. A green feedback box at the bottom of the question area says 'Good job! That's the correct answer.' At the bottom of the main area are 'Previous' and 'Next' buttons.

Google Cloud Skills Boost

← Main menu

Google Cloud Computing Foundations: Cloud Computing Fundamentals > Use Google Cloud to build your apps

Quiz

Your score: 83% Passing score: 80% Retake

Congratulations! You passed this assessment.

✓ 1. What is the Compute Engine feature that allows VMs to be added to or subtracted from an application based on load metrics?

☐ Network time protocol (NTP)

✓ ☒ Autoscaling

☐ Persistent disks

☐ Load balancing

Good job! That's the correct answer.

← Previous Next →

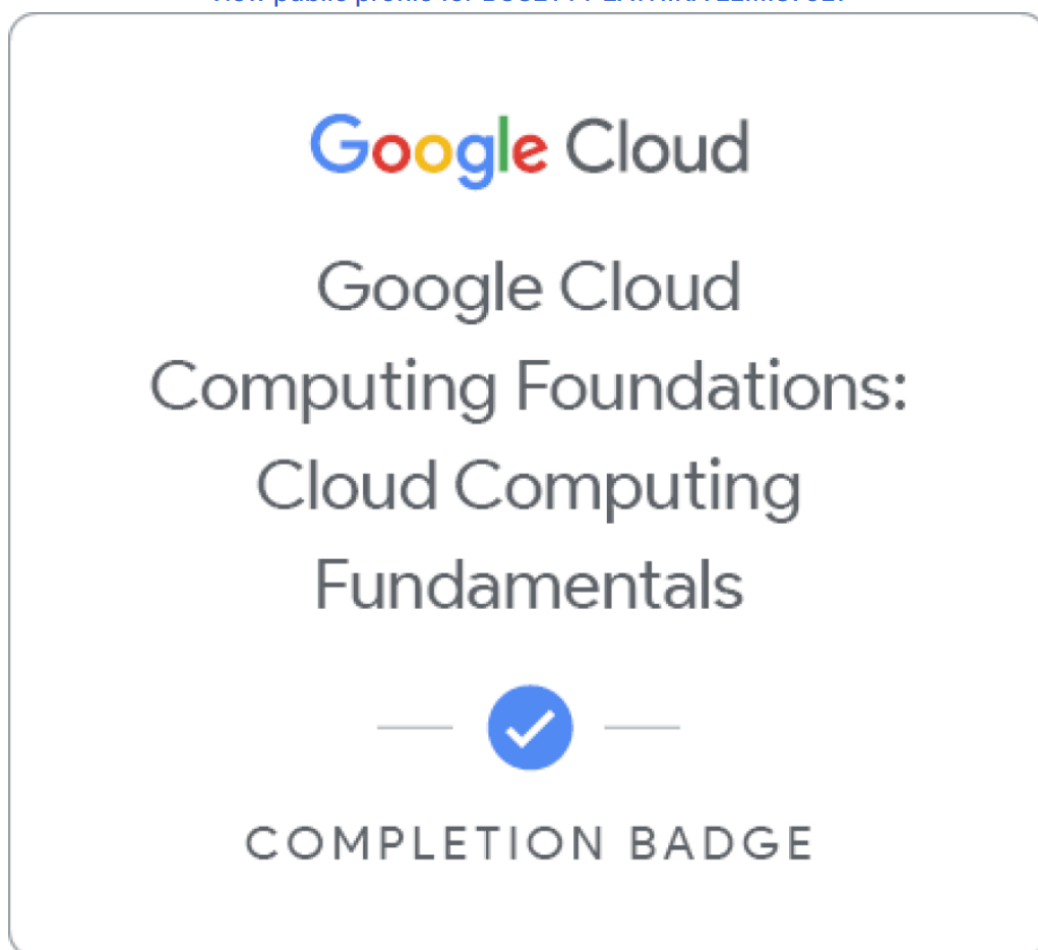
Course Summary:

The **Google Cloud Computing Foundations - Cloud Computing Fundamentals** course introduces learners to the essential concepts of cloud computing, including its benefits over traditional IT infrastructure such as scalability, flexibility, and cost-efficiency through pay-as-you-go pricing. It explains the different service models like Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS), using practical examples. The course also provides an overview of Google Cloud's global infrastructure and core services. Overall, it lays a strong foundation for anyone interested in exploring cloud technologies and pursuing a career in cloud computing.

Completion Badge:

BUSETTY LATHIKA 22MIS7027 has earned this award!

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Google Cloud Computing Foundations: Cloud Computing Fundamentals

Apr 1, 2025

Course Platform: Google Cloud Skills Boost

Course2: Google Cloud Fundamentals: Core Infrastructure

Course hours: 5 hours

Course Type: Completion Badge

Introduction:

Google Cloud Fundamentals: Core Infrastructure is an introductory course designed to provide a strong foundation in the essential concepts and services of Google Cloud Platform (GCP). It is ideal for those new to cloud computing or looking to understand how GCP can help modernize infrastructure, build scalable applications, and manage data effectively.

Through interactive lessons and hands-on labs, learners will explore core components such as virtual machines, storage options, networking, and IAM (Identity and Access Management). The course also touches on the principles of cloud computing and how GCP supports businesses in achieving reliability, scalability, and cost-efficiency.

By the end of this course, learners will understand the basic architecture of Google Cloud, key services like Compute Engine, Cloud Storage, and App Engine, and how to use GCP's web console and CLI tools. They'll gain the ability to make informed decisions about resource deployment, security policies, and cost management. This foundational knowledge prepares learners to progress into more advanced cloud engineering and data engineering courses, enabling them to build cloud-native applications or manage enterprise-scale infrastructure.

Additionally, I wanted to:

- **Infrastructure & Networking:** Kubernetes Engine, VPC design, hybrid connectivity, shared VPCs, firewall rules, Cloud NAT, Cloud Interconnect, peering setup, and DNS configuration.
- **Compute & Scaling:** Compute Engine, preemptible VMs, managed instance groups, autoscaling, custom images, startup scripts, and multi-region vs. zonal architecture.
- **Storage & Data:** Persistent disks, bucket lifecycle, CDN integration, Dataflow basics, Big Query analytics, and data locality principles.
- **Monitoring & Security:** Cloud Monitoring, Stack driver, logging sinks, metrics explorer, alert policies, IAM roles, service accounts, and SSH key management.
- **DevOps & Cost Management:** CI/CD pipelines, API management, gcloud SDK, resource hierarchy, logging strategies, billing export, budget alerts, cost optimization, failover planning, Cloud Run, Cloud Functions, and troubleshooting GCP services.

Choosing this course to build a strong foundation in Google Cloud and gain hands-on experience with core services essential for modern cloud infrastructure. It aligns perfectly with my goal of mastering cloud technologies for scalable, secure, and efficient solutions.

Module-Wise Analysis of the Course Content:

No. of Modules: 6

Module1

Introducing Google Cloud

Objective

The objective of Introducing Google Cloud is to familiarize learners with the core concepts, benefits, and global infrastructure of Google Cloud Platform. It aims to explain how GCP supports businesses with scalable, secure, and flexible cloud solutions.

Through this introduction, learners will explore key services like Compute Engine, App Engine, and Cloud Storage, while understanding GCP's resource hierarchy, billing structure, and identity management tools. This sets the stage for deeper cloud learning and practical application.

Lesson1: overview of cloud computing

Objective

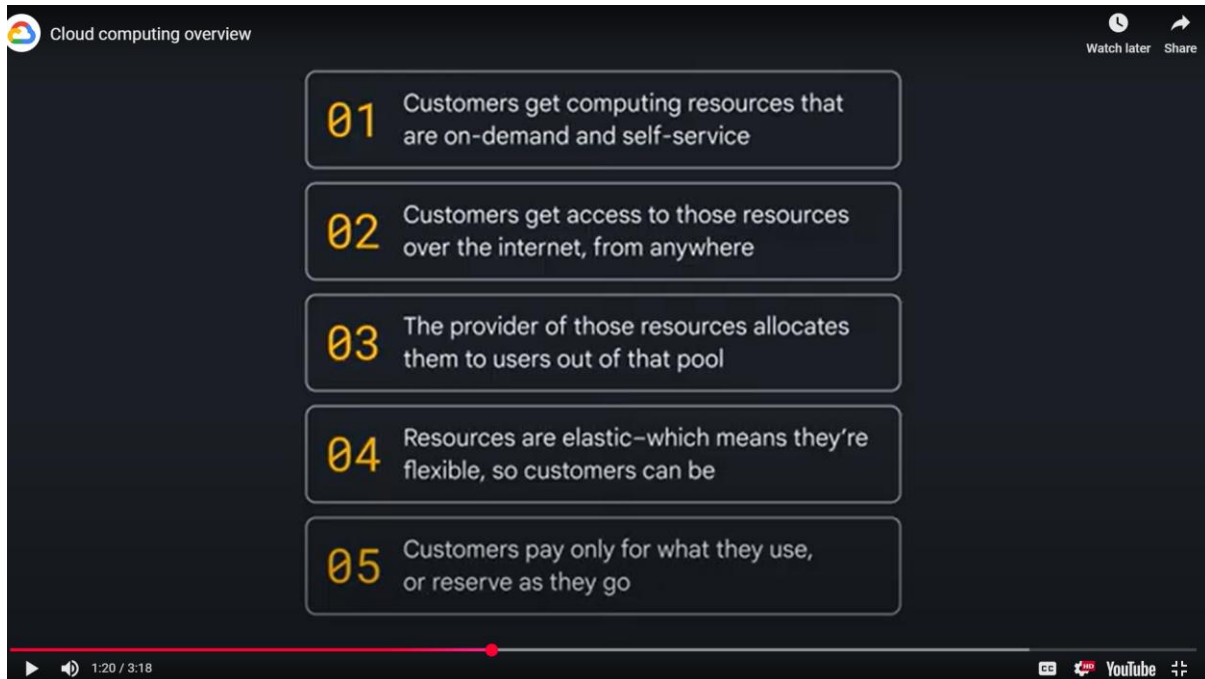
The objective of the Overview of Cloud Computing is to introduce the fundamental principles, models, and benefits of cloud technology. It aims to explain how cloud computing enables on-demand access to computing resources with scalability, flexibility, and cost-efficiency. This section also highlights the differences between IaaS, PaaS, and SaaS models.

Key Concepts

- On-Demand Self-Service – Users can provision computing resources like servers and storage automatically, without human interaction with service providers.
- Scalability and Elasticity – Cloud platforms allow resources to scale up or down dynamically based on demand, ensuring optimal performance and cost-efficiency.
- Service Models (IaaS, PaaS, SaaS) – Different levels of cloud services cater to varied needs, from full infrastructure control (IaaS) to ready-to-use applications (SaaS).

My Learning

I learned that cloud computing provides scalable, flexible, and cost-effective access to IT resources over the internet. I now understand the core service models—Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS)—and how each serves different business needs. Additionally, I gained insights into key cloud characteristics like on-demand provisioning, elasticity, and resource pooling, which make cloud environments efficient and user-centric.



Lesson2: IaaS and PaaS

Objective

The objective of IaaS (Infrastructure as a Service) is to provide virtualized computing resources like servers, storage, and networking on demand, giving users full control over their infrastructure. PaaS (Platform as a Service) aims to offer a managed environment for developing, testing, and deploying applications without worrying about underlying hardware or system software.

Key Concepts

- IaaS (Infrastructure as a Service) – Provides virtual machines, storage, and networking; users manage the OS and applications while the provider manages the infrastructure.
- PaaS (Platform as a Service) – Offers a ready-to-use platform for application development and deployment, abstracting away infrastructure management.
- Flexibility vs. Simplicity – IaaS offers greater control and customization, while PaaS simplifies development by managing runtime, scaling, and environment setup.

My Learning

In learned about IaaS and PaaS, I understood the differences in control, flexibility, and use cases between the two models. IaaS gives me the ability to manage and configure

infrastructure components, while PaaS simplifies the development process by handling the backend environment. This knowledge helps me choose the right model based on project needs and resource management.

Google Cloud logo IaaS and PaaS Watch later Share



Two new types of offerings:

- ✓ IaaS - infrastructure as a service
- ✓ PaaS - Platform as a service

0:10 / 2:29 CC BY-ND YouTube

Google Cloud logo IaaS and PaaS Watch later Share

IaaS model

Pay for what they allocate



PaaS model

Pay for what they use

0:53 / 2:29 CC BY-ND YouTube

Lesson 3: Google Cloud network

Objective

The objective of Google Cloud Network is to provide a comprehensive understanding of how Google Cloud's networking solutions ensure scalability, security, and reliability for global applications. It aims to teach learners how to design and manage network resources effectively across multiple regions, while leveraging Google Cloud's high-performance infrastructure.

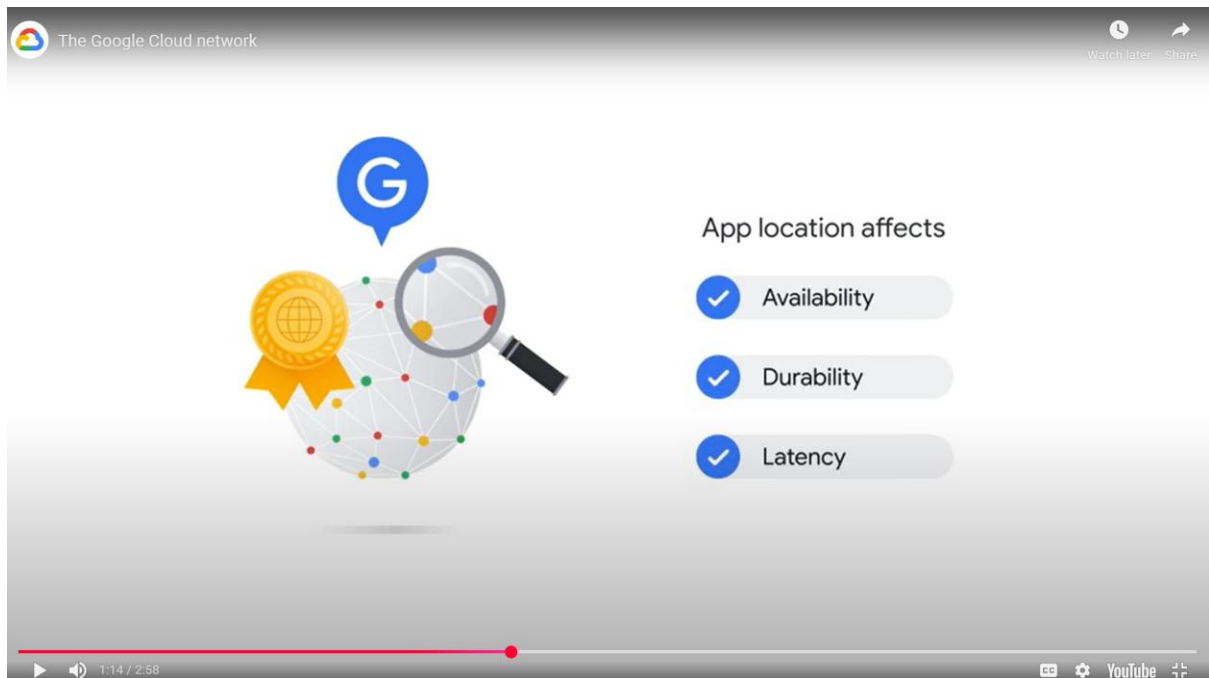
Through this, learners will explore key networking services like Virtual Private Cloud (VPC), Cloud Load Balancing, Cloud Interconnect, and DNS. The course also emphasizes secure network architecture and best practices for optimizing data flow and connectivity in a cloud environment.

Key Concepts

- Virtual Private Cloud (VPC) – A secure, isolated network within Google Cloud that allows users to define their own IP address ranges, subnets, and routing policies.
- Cloud Load Balancing – A fully distributed, software-defined load balancing solution that scales applications globally, balancing traffic across multiple instances.
- Cloud Interconnect – A service that connects on-premises infrastructure directly to Google Cloud, providing high-throughput, low-latency connectivity.
- DNS (Domain Name System) – Google Cloud DNS offers a globally distributed DNS service, enabling reliable and fast domain name resolution.
- Network Security – Tools like firewalls, private Google access, and Identity-Aware Proxy (IAP) to control and secure access to cloud resources.

My Learning

In learning about Google Cloud Network, I gained an understanding of how to build scalable, secure, and high-performance network architectures in the cloud. I learned to leverage tools like VPC, load balancing, and Cloud Interconnect for efficient connectivity and traffic management. This knowledge empowers me to design and optimize cloud networking solutions that ensure reliable and secure data flow across regions.



Lesson 4: Environmental Impact

Objective

The objective of Environmental Impact in accordance with Google Cloud is to highlight how Google Cloud implements sustainable practices to reduce the environmental footprint of its services. It aims to show how Google Cloud optimizes energy consumption, uses renewable energy, and builds green data centers to promote environmental sustainability while maintaining efficient, scalable cloud computing.

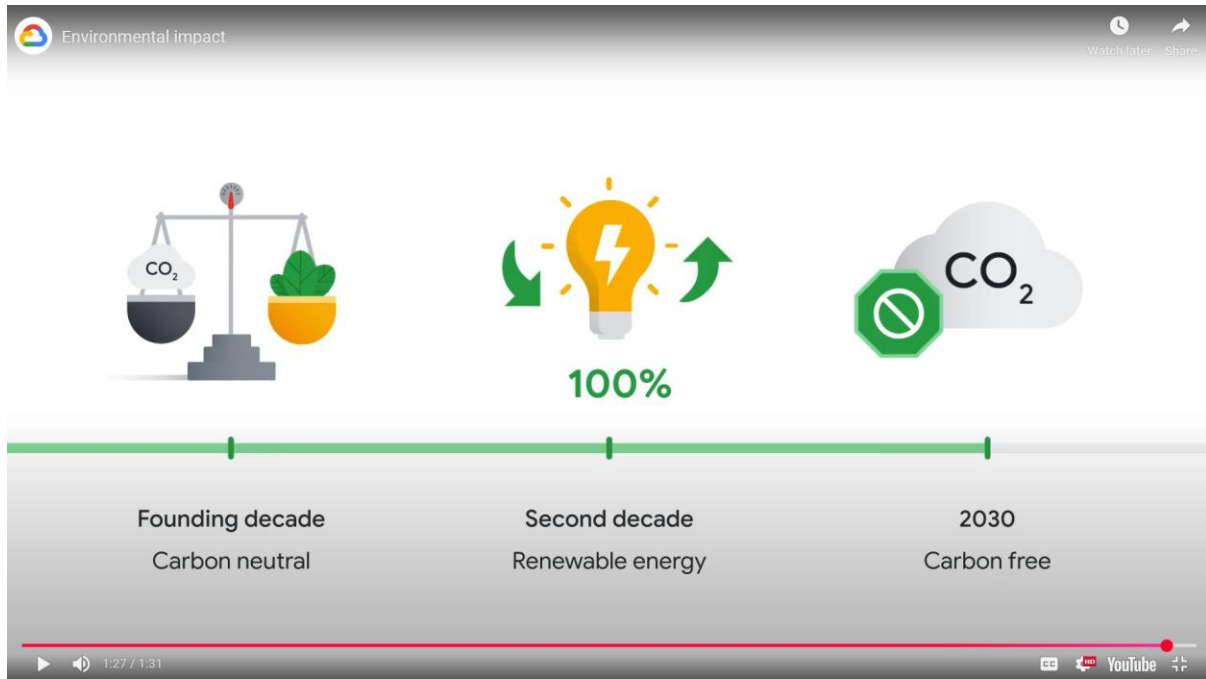
This course also focuses on the importance of cloud solutions in helping businesses reduce their carbon footprint by shifting to more energy-efficient and resource-efficient cloud infrastructures. The goal is to create awareness of how cloud computing can contribute to environmental goals and drive sustainable business practices.

Key Concepts

- **Energy Efficiency** – Cloud providers use advanced technologies and renewable energy to optimize energy consumption in data centers, reducing the environmental footprint.
- **Resource Sharing** – Cloud computing allows for resource pooling, which maximizes infrastructure use and minimizes waste, leading to more efficient energy consumption.
- **Green Data Centers** – Many cloud providers implement eco-friendly practices like cooling systems, renewable energy sources, and sustainable infrastructure design to lower carbon emissions.
- **Carbon Footprint Reduction** – By shifting to cloud services, businesses can reduce their on-premise energy use and associated carbon emissions through optimized cloud infrastructure.

My Learning

Through learning about the environmental impact of cloud computing, I gained insight into how cloud technologies contribute to energy efficiency and sustainability. I now understand how resource sharing and green data centers help reduce carbon footprints and promote eco-friendly practices in the industry.



Lesson 5: Security

Objective

The objective of Security in Google Cloud is to provide a comprehensive understanding of the tools and best practices available for securing data, applications, and services on Google Cloud. It aims to teach learners how to implement security measures like identity management, encryption, and access controls to safeguard cloud resources from threats.

Through this, learners will explore Google Cloud's security architecture, including its multi-layered security model, compliance with global standards, and built-in features like Cloud Identity and Access Management (IAM), Google Cloud Key Management, and security logging. This knowledge enables users to create secure, resilient cloud environments while maintaining compliance with industry regulations.

Key Concepts

- Identity and Access Management (IAM) – IAM helps control who can access specific resources and the level of permissions they have, ensuring secure access to cloud assets.

- Data Encryption – Google Cloud ensures data is encrypted at rest and in transit, protecting sensitive information from unauthorized access.
- Security Monitoring and Logging – Google Cloud provides tools like Cloud Security Command Center and Cloud Audit Logs to monitor and track security events, ensuring rapid detection and response to threats.

My Learning

Through learning about security in Google Cloud, I gained an understanding of how to effectively manage access control, encrypt data, and monitor security events. This knowledge enables me to implement robust security measures and protect cloud resources from potential threats and vulnerabilities.

Lesson 6: Open Source Ecosystem

Objective

The objective of the Open Source Ecosystem is to introduce learners to the benefits and challenges of open-source software in the context of modern computing. It aims to demonstrate how open-source communities collaborate to create software solutions that are freely accessible and customizable, driving innovation across industries.

This course also explores the key principles behind open-source development, such as transparency, community-driven contributions, and shared ownership. Learners will understand how these principles contribute to the rapid evolution of technologies and the widespread adoption of open-source tools.

Additionally, the course focuses on the integration of open-source software with cloud environments and how platforms like Google Cloud support and contribute to the open-source ecosystem. This knowledge equips learners to leverage open-source technologies for scalable, efficient, and cost-effective solutions.

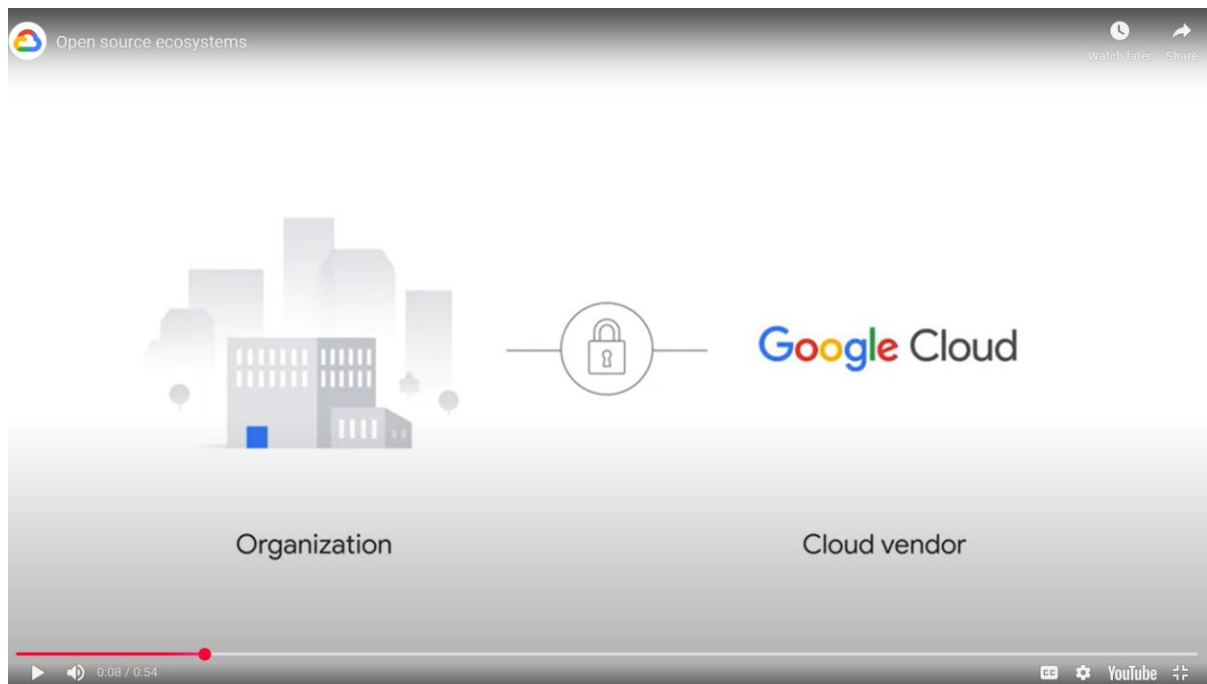
Key Concepts

- Community Collaboration – Open-source software is developed and maintained by a community of contributors who collaborate to improve the software, ensuring constant innovation and updates.
- Transparency and Customizability – Open-source projects allow users to view, modify, and distribute the source code, offering flexibility and customization to meet specific needs.
- Integration with Cloud Platforms – Many cloud providers, including Google Cloud, support open-source tools, enabling seamless integration and deployment of open-source technologies in cloud environments.

My Learning

In learning about the Open Source Ecosystem, I gained an understanding of how collaborative communities drive innovation and create freely accessible software. I learned how open-source principles like transparency and customizability allow for flexible, cost-

effective solutions, and how these technologies integrate seamlessly with cloud platforms to enhance scalability and efficiency.



Lesson 7: Pricing and Billing

Objective

The objective of Pricing and Billing in Google Cloud is to provide learners with an understanding of how Google Cloud's pricing model works and how to manage costs effectively. It focuses on explaining the different billing options, including pay-as-you-go, committed use, and sustained use discounts, helping businesses optimize their cloud spending.

The course also covers tools and strategies for monitoring and controlling cloud expenses, such as Google Cloud's Billing Console and budget alerts. Learners will gain skills to analyze cost reports, track usage, and make informed decisions to prevent overspending while maximizing resource efficiency.

Additionally, the course aims to teach how to estimate future cloud costs using the Pricing Calculator and implement cost optimization techniques. This knowledge ensures users can balance performance and cost-effectiveness when deploying cloud solutions, making Google Cloud an affordable option for businesses of all sizes.

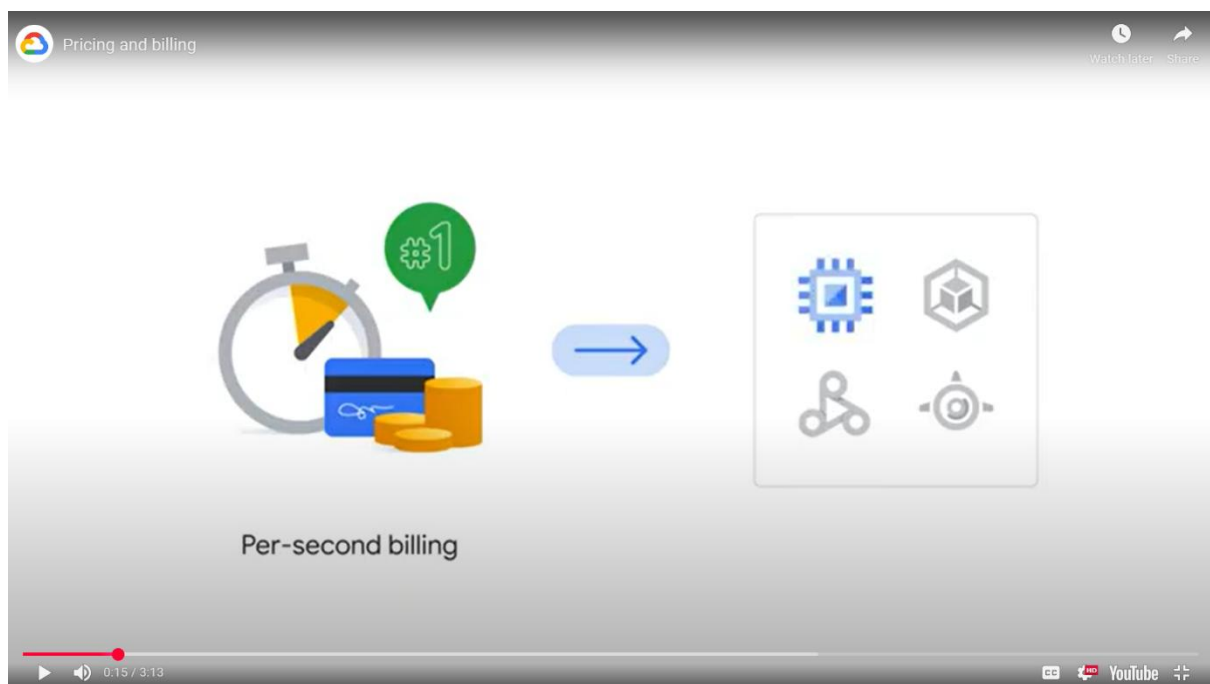
Key Concepts

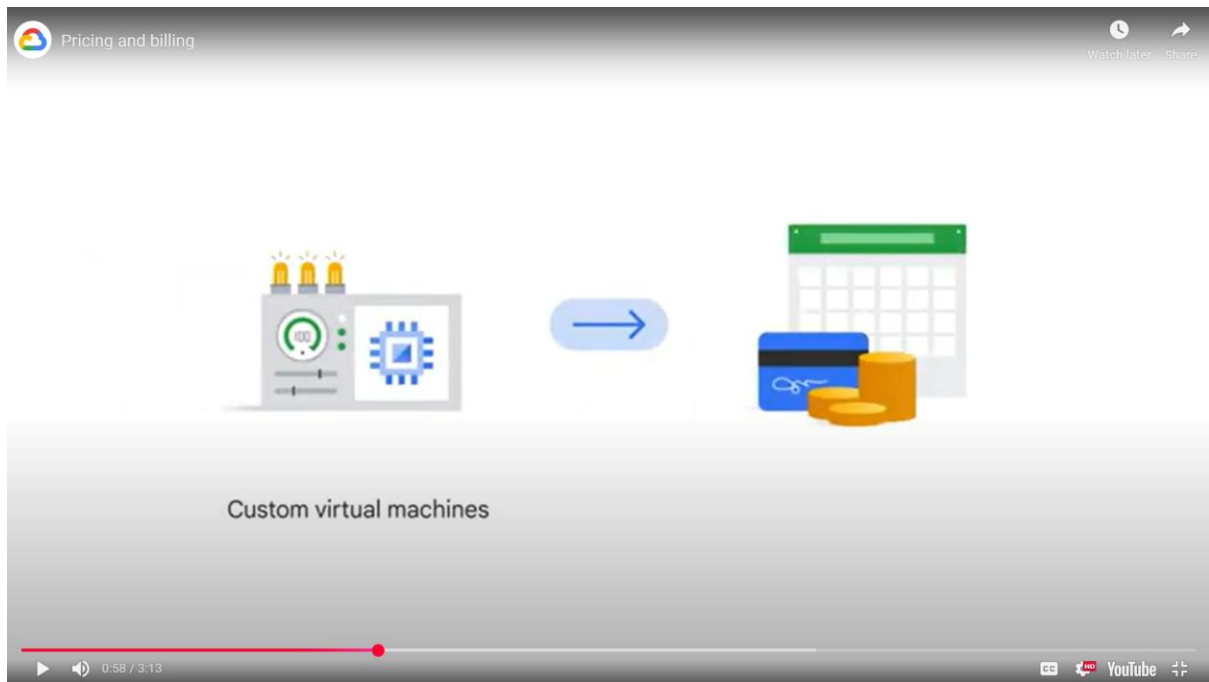
- Pay-as-you-go Pricing – Google Cloud charges based on actual usage, allowing businesses to only pay for the resources they consume, making it cost-effective and flexible.

- Sustained and Committed Use Discounts – Discounts are offered for long-term use or for committing to use specific services over a period, helping reduce overall cloud costs.
- Billing Console and Budget Alerts – Tools like the Billing Console and budget alerts help monitor cloud spending, track usage, and set cost thresholds to prevent overage charges.
- Pricing Calculator – Google Cloud's Pricing Calculator allows users to estimate costs based on their resource usage, providing a clear understanding of potential expenses.

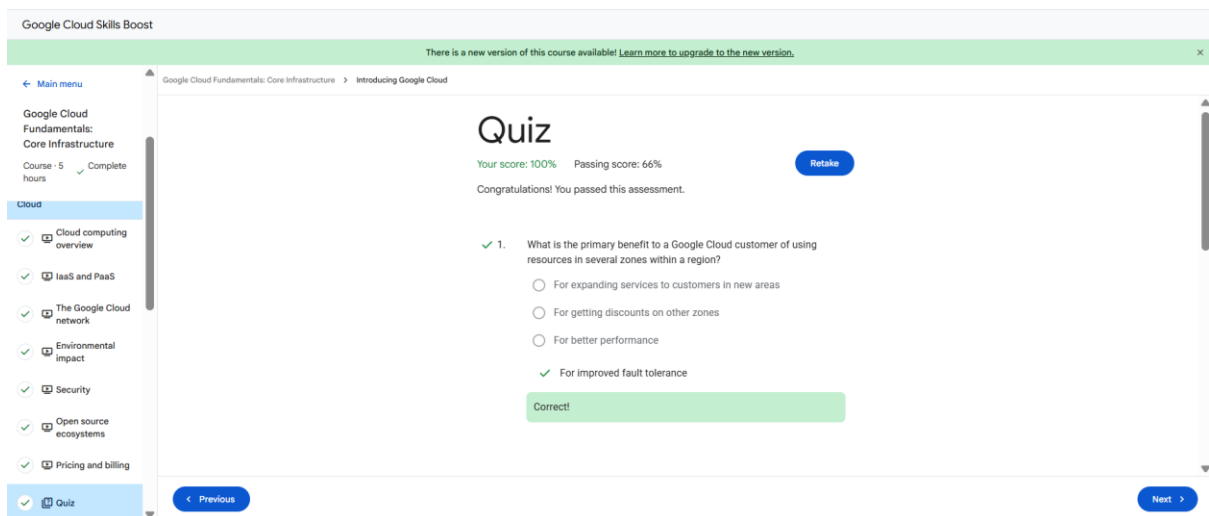
My Learning

Through learning about pricing and billing in Google Cloud, I gained a solid understanding of how Google Cloud's pricing models work, including pay-as-you-go, sustained use, and committed use discounts. I also learned how to use tools like the Billing Console and Pricing Calculator to track and manage cloud expenses effectively, helping optimize costs for cloud projects.





Lesson 8: Quiz



Module2

Resources and Access in the Cloud

Objective

The objective of Resources and Access in the Cloud is to help learners understand how to effectively manage and control cloud resources, including compute, storage, and networking. It focuses on teaching the best practices for provisioning, configuring, and managing cloud infrastructure to meet business needs.

This course also covers cloud access management, including Identity and Access Management (IAM), roles, and permissions, enabling secure and controlled access to cloud resources. Learners will gain insights into how to create and manage users, roles, and policies to maintain proper security and compliance.

Furthermore, the course explores the concept of resource organization and hierarchy, helping learners manage large-scale deployments by structuring projects, folders, and organizations. This knowledge equips users to efficiently manage resources while maintaining security, scalability, and cost-efficiency in a cloud environment.

Lesson1: Google Cloud's Resource Hierarchy

Objective

The objective of Google Cloud's Resource Hierarchy is to provide a structured approach to organizing and managing cloud resources in a scalable and secure manner. It helps users understand the relationship between organizations, folders, projects, and resources within Google Cloud.

This structure allows for centralized management of access control, billing, and policies across various teams and departments. By assigning IAM roles and applying policies at different levels of the hierarchy, organizations can ensure consistent governance and security.

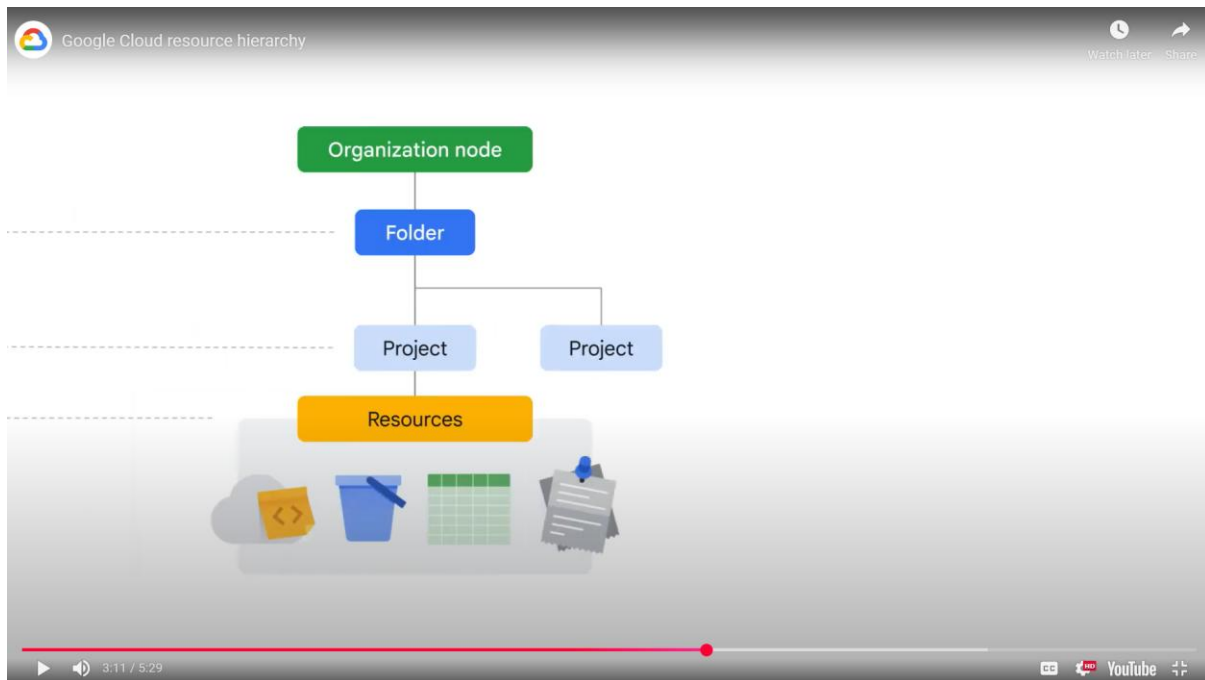
Additionally, it supports efficient resource allocation, easier collaboration, and simplified auditing, making it essential for managing large-scale cloud environments effectively and with control.

Key Concepts

- **Organization Node** – The top-level container that represents a company and allows centralized management of resources, IAM, and policies.
- **Folders and Projects** – Folders group related projects, while projects are the base-level containers for resources, services, and billing.
- **Policy Inheritance** – IAM roles and policies set at higher levels (like organizations or folders) automatically apply to lower levels, ensuring consistent access control.

My Learning

From learning about Google Cloud's Resource Hierarchy, I understood how to structure and organize resources using organization nodes, folders, and projects. I also learned how policy inheritance and IAM role management help ensure secure, scalable, and efficient control over access and cloud operations across different levels of the hierarchy.



Lesson2: Identity and Access Management (IAM)

Objective

The objective of Identity and Access Management (IAM) in Google Cloud is to provide secure and granular control over who can access specific cloud resources and what actions they can perform. It ensures that the right individuals or services have the appropriate permissions to perform their tasks while maintaining strong security boundaries.

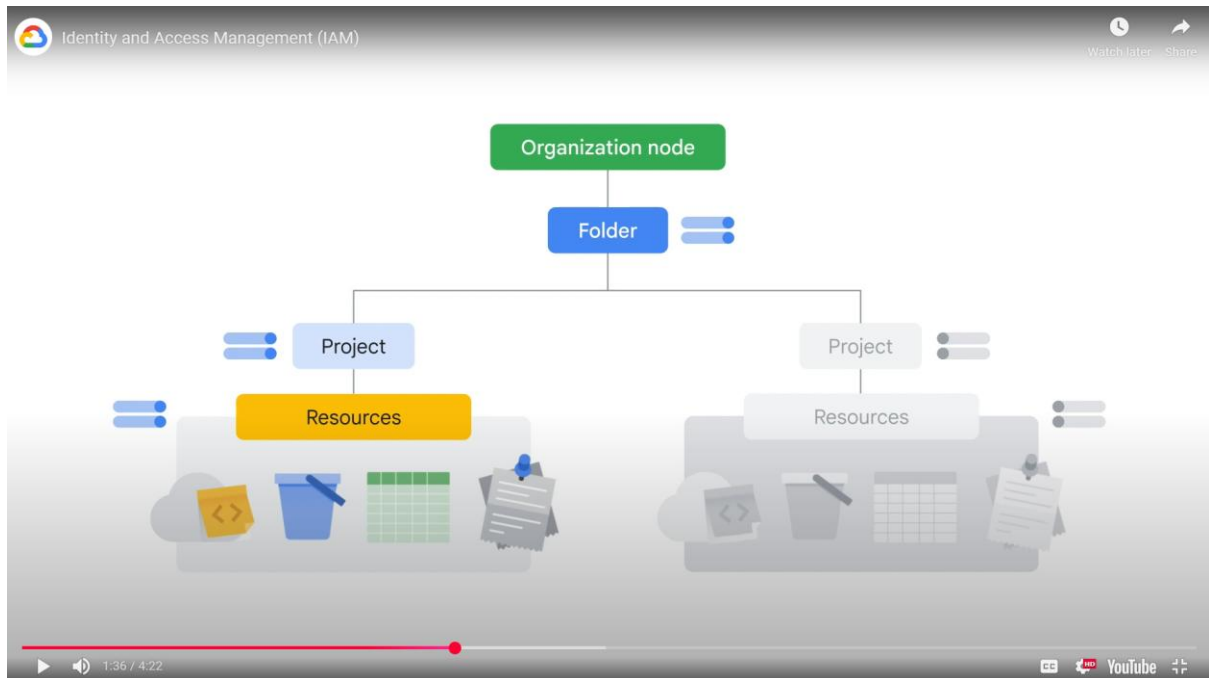
IAM also enables organizations to apply the principle of least privilege by assigning only necessary roles, reducing the risk of unauthorized access. With centralized management, it helps enforce consistent access policies, audit actions, and meet compliance requirements effectively.

Key Concepts

- Roles and Permissions – IAM uses predefined, custom, and basic roles to grant specific permissions to users, groups, or service accounts for fine-grained access control.
- Principals and Policies – IAM policies bind principals (like users or service accounts) to roles, defining what access is granted to which resources.
- Least Privilege Principle – Access should only be given to the resources and actions necessary for a user's role, minimizing potential security risks.

My Learning

From learning about Identity and Access Management, I understood how to control access to Google Cloud resources using roles, policies, and service accounts. I also learned the importance of applying the principle of least privilege to enhance security and manage permissions effectively.



Lesson 3: Service Accounts

Objective

The objective of Service Accounts in Google Cloud is to enable secure, automated access to cloud resources by applications, virtual machines, and other services. Unlike user accounts, service accounts represent non-human users that need to authenticate and interact with Google Cloud APIs and services programmatically.

Service accounts allow developers and administrators to assign specific roles and permissions tailored to the task that the application or service needs to perform. This ensures a clear separation of access rights and helps follow security best practices, such as the principle of least privilege.

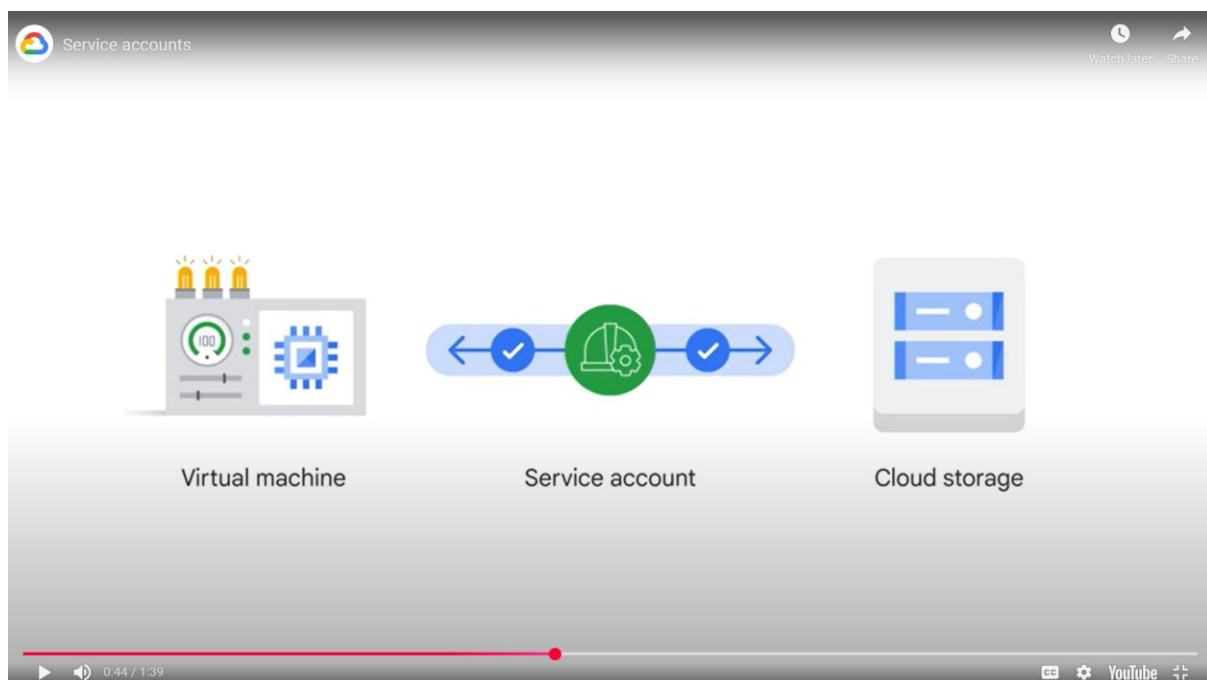
Additionally, service accounts support secure authentication methods using keys or built-in identity tokens. They can be customized, monitored, and rotated as needed, allowing for fine-grained access control and easy integration into CI/CD pipelines, automation scripts, and other cloud-native workflows.

Key Concepts

- Authentication for Non-Human Users – Service accounts allow applications, virtual machines, and services to authenticate with Google Cloud APIs and access resources securely.
- Role-Based Access – Permissions for service accounts are granted through IAM roles, ensuring that each service account has the appropriate level of access for its function.
- Key-Based Authentication – Service accounts use private keys or identity tokens for authentication, providing secure and automated access to cloud resources.
- Least Privilege and Monitoring – Service accounts follow the principle of least privilege by granting only necessary permissions and can be monitored for usage and activity.

My Learning

From learning about Service Accounts, I understood how they enable secure, automated access to Google Cloud resources for applications and services. I also learned how to assign specific IAM roles to these accounts, ensuring proper access control while maintaining security through key-based authentication.



Lesson 4: Cloud Identity

Objective

The objective of Cloud Identity is to provide a comprehensive identity management solution for organizations using Google Cloud. It allows businesses to manage users, groups, and devices with centralized control while integrating seamlessly with Google Workspace and other Google Cloud services.

Cloud Identity enables secure single sign-on (SSO) capabilities, simplifying user access across multiple applications and services. It also supports multi-factor authentication (MFA) to enhance security, ensuring that only authorized users can access critical resources.

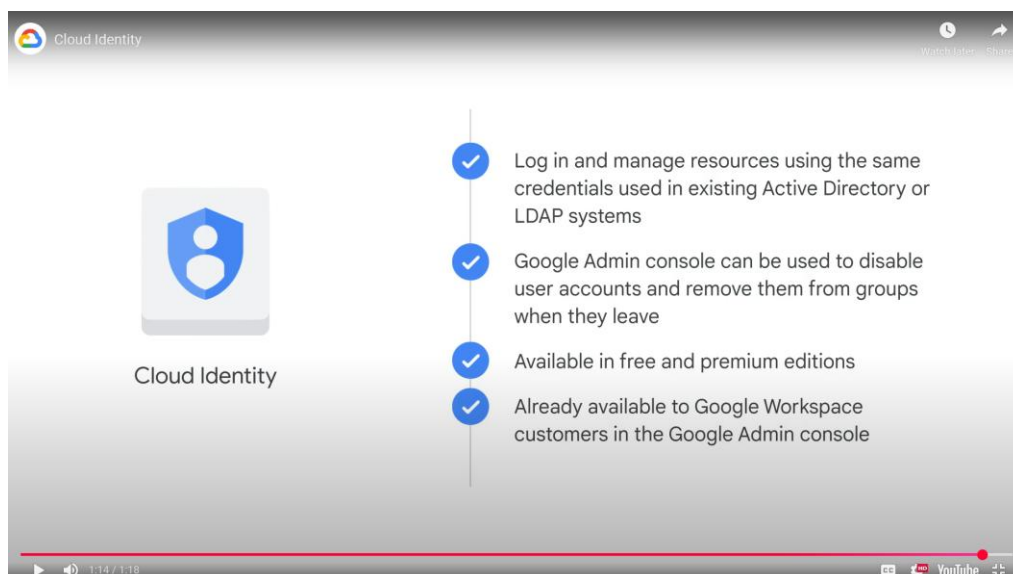
Additionally, Cloud Identity offers tools for managing devices, user provisioning, and access controls. It streamlines the administration of users and devices within an organization, ensuring compliance and security while enabling efficient collaboration across teams and services.

Key Concepts

- **User and Group Management** – Cloud Identity allows administrators to create, manage, and organize users and groups, simplifying access control and collaboration within an organization.
- **Single Sign-On (SSO)** – It enables Single Sign-On across various cloud applications, reducing the need for multiple logins and improving user experience while maintaining security.
- **Multi-Factor Authentication (MFA)** – Cloud Identity supports MFA, adding an additional layer of security by requiring users to verify their identity with multiple methods beyond just passwords.
- **Device Management** – Administrators can manage and secure devices that access cloud resources, ensuring that only authorized devices are used to access sensitive data.
- **Directory Integration** – Cloud Identity integrates with other directory services, allowing businesses to extend existing identity management systems and seamlessly integrate with on-premise solutions.

My Learning

From learning about Cloud Identity, I gained an understanding of how to manage users, groups, and devices securely while integrating single sign-on and multi-factor authentication. I also learned how it simplifies access control and enhances security across Google Cloud services and applications.



Lesson 5: Interacting with Google Cloud

Objective

The objective of Interacting with Google Cloud is to provide learners with the skills to effectively interact with Google Cloud services using both the Cloud Console and command-line tools. It helps users understand how to navigate and utilize the Google Cloud Console for deploying and managing resources in a user-friendly interface.

Additionally, the course introduces the gcloud command-line tool, teaching users how to perform tasks like managing virtual machines, storage, and networks programmatically. This empowers users to automate their workflows and perform tasks more efficiently compared to using the Console alone.

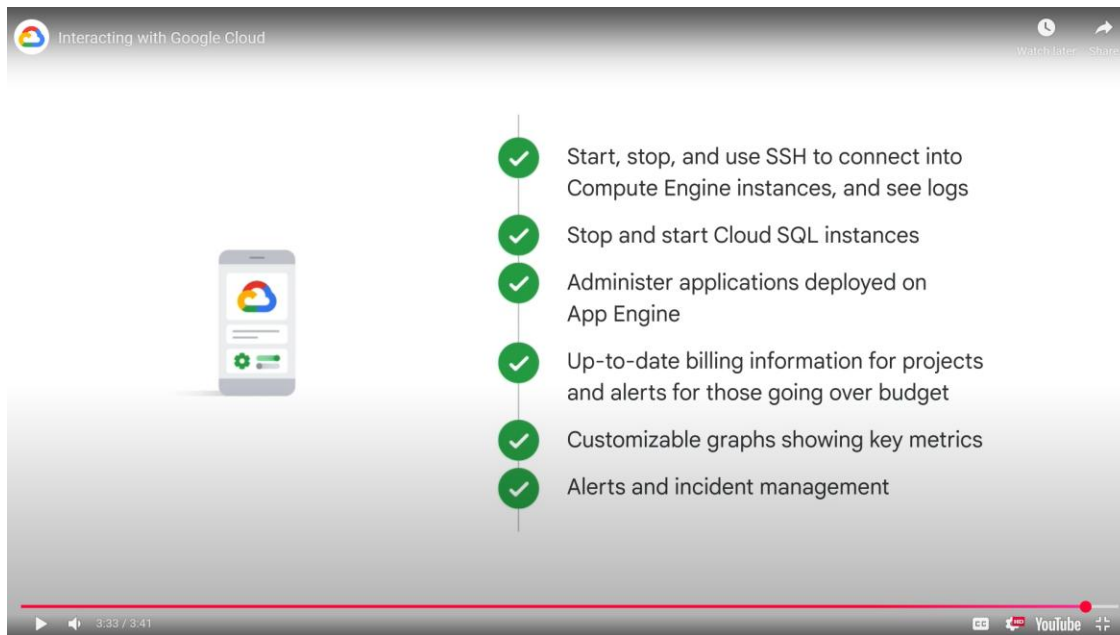
The course also covers best practices for interacting with cloud resources in a scalable and secure manner, ensuring that users can efficiently manage their cloud infrastructure, troubleshoot issues, and implement changes across multiple services within Google Cloud.

Key Concepts

- Cloud Console – The Google Cloud Console provides a graphical user interface to manage resources, configure services, and monitor cloud activities in a user-friendly environment.
- gcloud Command-Line Tool – The gcloud CLI allows users to interact with Google Cloud services programmatically, enabling efficient management of resources and automation of tasks.
- Cloud SDK – Google Cloud SDK is a collection of tools that include the gcloud CLI, allowing users to manage resources from local machines, integrate with CI/CD pipelines, and perform bulk operations.
- APIs and Cloud Client Libraries – Interacting with Google Cloud via RESTful APIs and client libraries enables developers to automate tasks and integrate Google Cloud services into their applications.

My Learning

From interacting with Google Cloud, I learned how to manage and deploy cloud resources using both the Cloud Console and the gcloud CLI tool. I also gained experience in automating tasks and efficiently navigating services through the Cloud SDK and client libraries.



Lab1: Getting Started with Cloud Marketplace

Duration Time: 25 minutes

Objective:

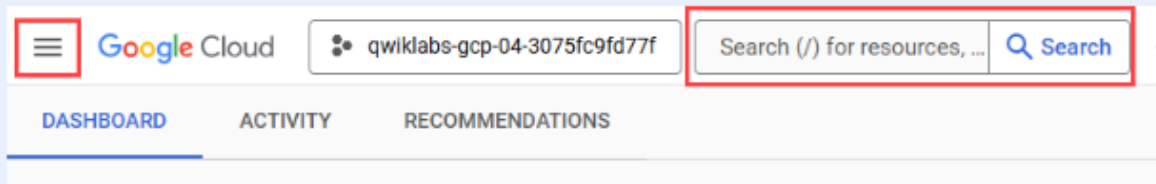
The objective of Getting Started with Cloud Marketplace is to introduce users to the wide range of third-party solutions and Google Cloud services available for quick deployment through the Google Cloud Marketplace. It aims to simplify the process of discovering, launching, and managing applications directly within the Google Cloud ecosystem.

Steps:

Task 1: Sign in to the Google Cloud Console

- Click the Start Lab button
- If prompted for payment, follow the instructions to proceed.
- View Lab Details panel on the left side:
- Click Open Google Cloud Console (opens in a new tab).
- Copy the Username and Password provided.
- Sign in with temporary credentials:
- Paste the Username and click Next.
- Paste the Password and click Next.
- Accept terms and continue:
- Accept terms and conditions.
- Do not set up recovery options or 2FA.
- Do not sign up for free trials.
- Wait for the Google Cloud Console to load.
- Use the Navigation Menu (≡) to access services.

Note: To view a menu with a list of Google Cloud products and services, click the **Navigation menu** at the top-left, or type the service or product name in the **Search** field.



Task 2: Deploy a LAMP Stack Using Cloud Marketplace

- In the Google Cloud Console:
- Click the Navigation Menu (☰) → Marketplace.
- In the search bar, type: LAMP → Press Enter.
- Click on Bitnami package for LAMP (not Google Click to Deploy).
- On the LAMP stack page:
- Click GET STARTED.
- Accept the Terms and Agreements, then click AGREE.
- On the confirmation popup, click DEPLOY.
- If prompted, click Enable for:
- Compute Engine API
- Infrastructure Manager API
- In the deployment form:
- Set Zone to the one given in the lab instructions.
- Set Series to E2 and Machine Type to e2-medium.
- Leave all other settings at default.
- Click DEPLOY.
- If a “Welcome to Deployment Manager” message appears, click Close.
- Wait for the status to change from being deployed to has been deployed.

Task 3: Verify Your Deployment

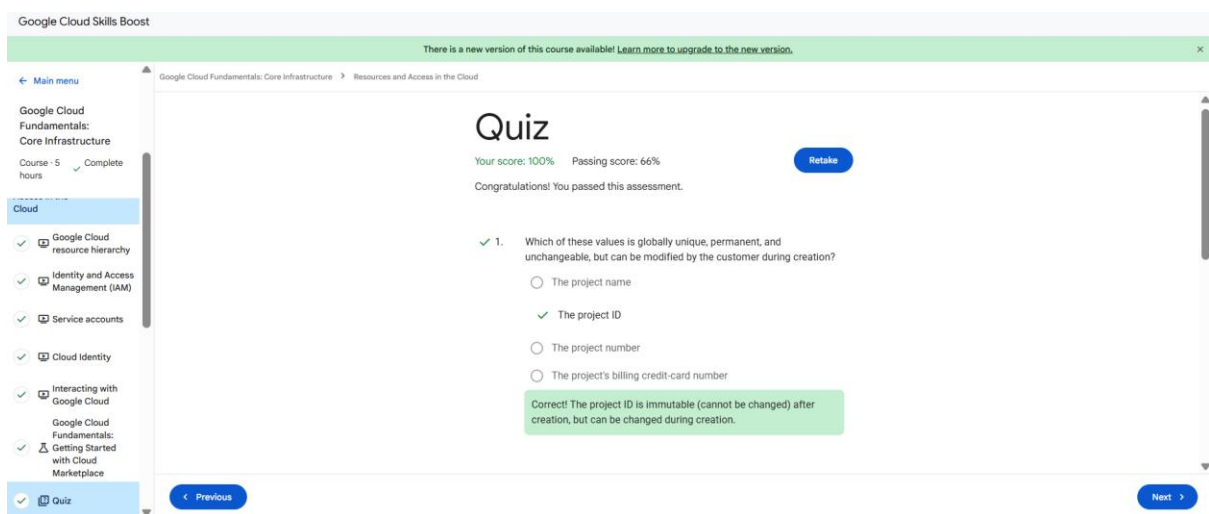
- When deployment is complete, go to:
- lampstack-1 > Details
- Click on the Site URL.
- If the site doesn't respond, wait 30 seconds and retry.
- Follow the redirection link if prompted.
- A new browser tab will display a congratulations page, confirming that the Apache HTTP Server is running.
- Troubleshooting tip:
If the page doesn't load:
- Exit any corporate VPN/network.

- Try opening the IP on another device (e.g., phone/tablet).

Congratulations!

In this lab, you deployed a LAMP stack to a Compute Engine instance.

Quiz



Module3

Virtual machines and network in the cloud

Objective:

The objective of the lesson on Virtual Machines and Networks in the Cloud is to provide learners with a comprehensive understanding of how Google Cloud manages virtualized infrastructure through Compute Engine and VPC networking. This section aims to help users effectively create and manage virtual machines while ensuring secure, scalable, and high-performing network architecture. It introduces the concept of virtual private cloud networking, which serves as a foundational element for isolating and organizing cloud resources efficiently.

Learners will explore the functionality of Compute Engine for provisioning virtual machines, enabling them to deploy workloads with customizable configurations. It also covers the principles and practices of scaling virtual machines based on demand, ensuring high

availability and resource optimization. Another key objective is to highlight important VPC compatibilities that support hybrid networking, peering, and shared VPC use cases.

Lab1: Getting Started with VPC Networking and Google Compute Engine

Duration Time: 40 minutes

Objective:

The objective of the Getting Started with VPC Networking and Google Compute Engine lab is to introduce learners to the foundational concepts and operations of Google Cloud's Virtual Private Cloud (VPC) networking and the Compute Engine service. Participants will explore how the default network functions, including its subnets, routes, and firewall rules. The lab demonstrates the impact of deleting the default network and how to recreate a custom VPC network using auto mode with similar configurations. Learners will create and manage virtual machine instances in multiple regions, understand how internal and external IP addresses are assigned, and test network connectivity between VMs using tools like SSH and ping. Additionally, the lab highlights the importance of firewall rules in managing access and controlling traffic flow, helping users gain practical experience in setting up and securing cloud-based virtual infrastructure.

Steps:

Task 1. Explore the default network

- Each Google Cloud project has a default network that includes subnets, routes, and firewall rules. To view the subnets, navigate to VPC network > VPC networks in the Cloud Console. Click on the default network, then Subnets. You will see a subnet in each Google Cloud region with an associated CIDR block and gateway.
- To view the routes, click Routes from the left pane. In Effective Routes, select the default network and your assigned Lab Region, then click View. Notice there is a route for each subnet. These are managed automatically, but custom static routes can be created to direct traffic, such as through a NAT gateway.
- Next, click Firewall in the left pane. You'll see four Ingress firewall rules: default-allow-icmp, default-allow-rdp, default-allow-ssh, and default-allow-internal. These allow specific traffic types internally and externally based on protocol and port configurations.
- Now, select and delete all default network firewall rules. Confirm deletion. Then go back to VPC networks and delete the default network. Confirm deletion and verify that no routes or firewall rules remain. Try to create a VM instance under Compute Engine > VM instances. Click Create instance and accept defaults. You will see an error indicating that a VPC network is required.

Task 2: Create a VPC network and VM instances

- To create a new network, go to VPC networks and click Create VPC network. Name it mynetwork, select Automatic for subnet creation, and enable all firewall rules. Click Create. When done, confirm that subnets were created in all regions. This network replicates the default network setup.

- Now, create a VM instance in Region 1 by going to Compute Engine > VM instances and clicking Create Instance. Name the instance mynet-us-vm, select Region 1 and Zone 1, choose the E2 series and the e2-micro machine type, and click Create.
- Repeat the process to create another instance named mynet-r2-vm in Region 2 and Zone 2 with the same machine configuration. Note that both instances are assigned ephemeral external IP addresses.

Task 3: Explore the connectivity for VM instances

- Check connectivity by SSHing into mynet-us-vm using the external IP. You can connect because the allow-ssh rule permits tcp:22 access. From within the VM, ping the internal and external IPs of mynet-r2-vm. You will see successful responses due to the allow-icmp and allow-custom rules.
- Next, delete the allow-icmp rule from the firewall and try pinging again. The external IP ping will now fail, while the internal IP ping still works due to the allow-custom rule.
- Then delete the allow-custom rule. Both internal and external IP pings should now fail, confirming that firewall rules govern this connectivity.
- Finally, delete the allow-ssh rule and attempt to SSH again to mynet-us-vm. The connection will fail, confirming SSH access is blocked without the rule.

Task 4: Review

- In this lab, we examined the default VPC network's components, including subnets, routes, and firewall rules. After deleting the default network, you confirmed that VM creation is not possible without a VPC. You then created a new auto mode network and deployed two VM instances. We tested VM connectivity and observed how removing firewall rules affects access via ICMP and SSH. This exercise reinforced the critical role of firewall rules and VPC networking in managing virtual machine communication.

Quiz

The screenshot shows a quiz interface for Google Cloud Fundamentals: Core Infrastructure. The quiz is titled "Quiz" and shows a score of 100% (Your score: 100%, Passing score: 80%). A "Retake" button is visible. The question is: "1. In Google Cloud VPCs, what scope do subnets have?" The options are Global, Zonal, Multi-regional, and Regional. The correct answer is Regional, which is highlighted with a green checkmark. A feedback message states: "That's correct. VPC subnets can span the zones that make up a region. This is beneficial because your solutions can incorporate fault tolerance without complicating your network topology." The interface includes a sidebar with a "Main menu" and a list of topics: Compute engine, Scaling virtual machines, Important VPC compatibilities, Cloud Load Balancing, Cloud DNS and Cloud CDN, Connecting networks to Google VPC, Getting Started with VPC, and Networking and Google Compute Engine. The "Quiz" item is selected in the sidebar. Navigation buttons for "Previous" and "Next" are at the bottom.

Module 4

Storage in the Cloud

Objective:

To gain a foundational understanding of Google Cloud's diverse storage solutions and learn how to leverage them for different types of data management. This includes exploring Google Cloud Storage for object storage, understanding Cloud SQL, Cloud Spanner, Firestore, and Bigtable for relational and NoSQL database needs, and learning about storage classes and data transfer methods to optimize performance and cost. The course will also guide you through comparing different storage options and provide practical insights to get started with these technologies, with a final quiz to test your knowledge.

Lab1: Google Cloud Fundamentals: Getting Started with Cloud Storage and Cloud SQL

Duration Time: 1hr 15 minutes

Objective:

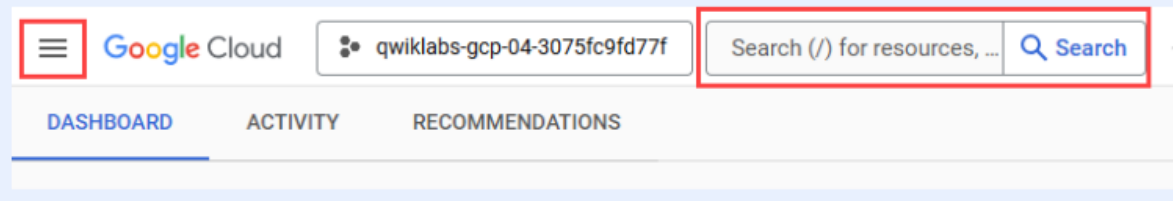
Introduce learners to the fundamentals of using Cloud Storage and Cloud SQL on Google Cloud. This lab guides users through the process of creating a Cloud Storage bucket, uploading and managing objects, and setting up a Cloud SQL instance. It provides hands-on experience with storing data in the cloud and accessing it securely, helping learners understand basic storage and database operations within the Google Cloud environment.

Steps:

Task 1: Sign in to the Google Cloud Console

- Start the lab and note the temporary credentials provided.
- Open the Google Cloud Console using the provided link.
- Sign in using the lab credentials (not your own account).
- Accept terms, skip recovery options and free trial.
- Console opens with access to Google Cloud services.

Note: To view a menu with a list of Google Cloud products and services, click the **Navigation menu** at the top-left, or type the service or product name in the **Search** field.



Task 2: Deploy a Web Server VM Instance

- Go to Compute Engine > VM Instances and create a new instance named "bloghost".
- Choose the assigned region and zone.
- Use the default machine type and Debian GNU/Linux 12 (bookworm) image.
- Enable HTTP traffic in firewall settings.
- Add a startup script to install Apache, PHP, and MySQL PHP libraries.
- Save the internal and external IPs of the instance for later use.

Task 3: Create a Cloud Storage Bucket Using gcloud

- Launch Cloud Shell and set your location (US, EU, or ASIA).
- Create a bucket named after your project ID.
- Copy an image from a public Cloud Storage location to your bucket.
- Make the image publicly accessible by modifying its Access Control List.

Task 4: Create the Cloud SQL Instance

- Go to SQL and create a MySQL instance named "blog-db".
- Select the "Sandbox" edition and your assigned region and zone.
- Set a secure password for the root account.
- After deployment, copy the instance's public IP address.
- Add a new user account named "blogdbuser" with a custom password.
- Allow access to the database from your VM's external IP by configuring the network settings.

35.192.208.2/32

Task 5: Configure the Web Server to Connect to Cloud SQL

- SSH into your VM instance.
- Navigate to the web server's root directory.
- Create and edit an index.php file to include a connection script to the SQL instance.

- Initially, load the web page to confirm connection error appears.
- Edit the file again to replace placeholders with actual Cloud SQL IP and password.
- Reload the web page to confirm a successful database connection message appears.

```
35.192.208.2/index.php
```

```
Database connection failed: ...
```

```
Connected successfully
```

Task 6: Add a Cloud Storage Image to the Web Page

- In Cloud Storage, find the public URL of the uploaded image.
- Edit the index.php file on the VM to include an HTML tag referencing the image URL.
- Save the file and restart the web server.
- Reload the web page to see the banner image displayed above the blog title.

```
<img src='https://storage.googleapis.com/qwiklabs-gcp-0005e186fa559a09/my-excellent-blog.png'>
```

Congratulations!

In this lab, you configured a Cloud SQL instance and connected an application in a Compute Engine instance to it. You also worked with a Cloud Storage bucket.

Module 5

Applications in the Cloud

Objective:

To understand how modern applications are developed, deployed, and scaled using cloud-native technologies on Google Cloud, with a focus on leveraging services like App Engine, Cloud Functions, and Kubernetes Engine to build resilient, scalable, and cost-effective solutions.

Lesson 1: Cloud Run

Objective:

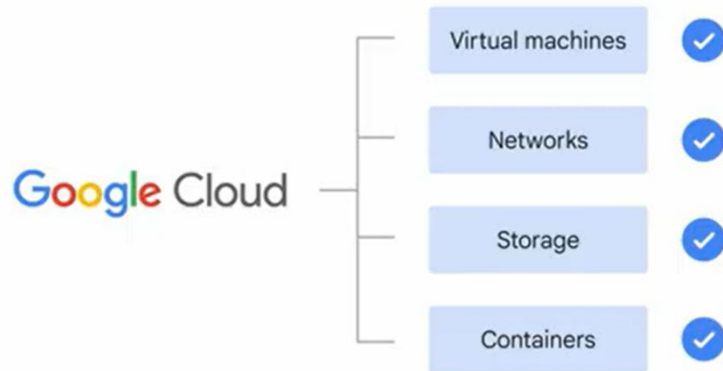
To introduce learners to Cloud Run, a fully managed compute platform that automatically scales containerized applications. The goal is to help learners understand how to deploy, manage, and scale applications without worrying about server infrastructure, while taking advantage of the flexibility of containers and the simplicity of serverless computing.

Key Concepts:

- What is Cloud Run?
- A fully managed serverless platform that runs containerized applications.
- Based on Native, built for portability and scalability.
- Containers in Cloud Run:
 - Packages code and dependencies into a container.
 - Supports any language, library, or binary that can run in a container.
- Autoscaling:
 - Automatically scales up to handle traffic spikes.
 - Scales down to zero when not in use (cost-efficient).
- Stateless Services:
 - Cloud Run is ideal for stateless workloads.
 - Designed to respond quickly to HTTP requests.
- HTTP-based Communication:
 - Services are exposed via HTTPS endpoints.
 - Can be triggered through webhooks, mobile apps, APIs, etc.
- Deployment Simplicity:
 - Deploy with a single command using the gcloud CLI.
 - Supports source-based and container-based deployment.
- Integration with Google Cloud Ecosystem:
 - Works seamlessly with Cloud Build, Cloud Storage, IAM, and Cloud Monitoring.

My Learning :

- Understood the core functionality and benefits of Cloud Run as a serverless platform.
- Learned how to deploy containerized applications easily without managing infrastructure.
- Gained insight into how Cloud Run scales automatically, making it ideal for variable workloads.
- Explored how to trigger Cloud Run services via HTTP requests and integrate with other Google Cloud services.
- Practiced using the gcloud CLI to deploy an app, configure permissions, and view service endpoints.
- Realized the cost and resource optimization Cloud Run provides, especially for on-demand services or microservices.
- Recognized use cases such as webhooks, APIs, event-driven architectures, and backend processing.



Lesson 2: Development in the Cloud

Objective:

To introduce learners to cloud-based development tools and services offered by Google Cloud, enabling them to build, test, and deploy applications more efficiently. This lesson focuses on how the cloud accelerates the development lifecycle through managed services, APIs, and tools for continuous integration and delivery (CI/CD), fostering faster innovation and collaboration.

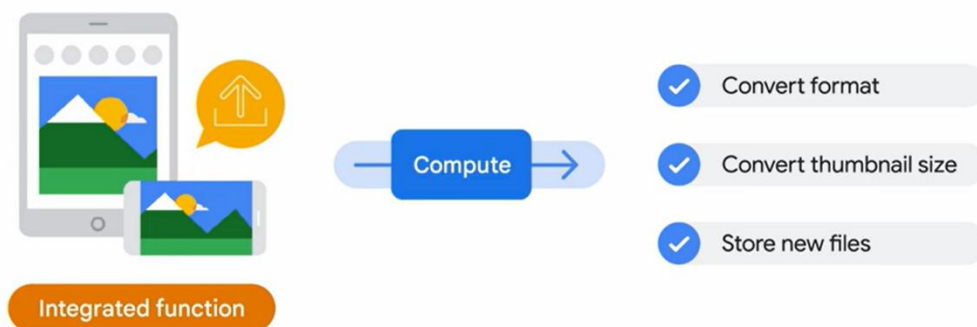
Key Concepts:

- Cloud-Based Development Environment:
- Develop applications directly in the cloud using tools like Cloud Shell, Cloud Code, and Cloud Source Repositories.
- APIs and Managed Services:
- Accelerate development by integrating with Google Cloud APIs (e.g., Vision API, Translate API).
- Rely on managed services to reduce operational overhead.
- Cloud SDK & Cloud Shell:
- Use Cloud SDK and Cloud Shell for a browser-based development experience without needing local setup.
- Built-in tools like Git, gcloud, and editors available instantly.
- Source Control and Collaboration:
- Use Cloud Source Repositories for Git-based version control.
- Support for collaboration among distributed teams.
- CI/CD in the Cloud:
- Use Cloud Build and Cloud Deploy for continuous integration and continuous delivery pipelines.

- Automate testing, building, and deployment of applications.
- Serverless Development:
- Leverage services like Cloud Functions and Cloud Run to deploy code without managing servers.

My Learning:

- Understood the benefits of developing applications directly in the cloud, including flexibility, speed, and accessibility.
- Learned how to use tools like Cloud Shell and Cloud Code for streamlined, browser-based coding and deployment.
- Gained familiarity with Cloud SDK and how it enables command-line interaction with Google Cloud services.
- Explored how APIs and managed services can be integrated easily into cloud-native apps to add powerful features quickly.
- Experienced the importance of CI/CD pipelines using Cloud Build and Cloud Deploy for automating the development workflow.
- Discovered how source control with Cloud Source Repositories supports collaboration and version tracking in distributed environments.
- Built confidence in creating scalable, serverless solutions while focusing on writing code, not managing infrastructure.



Lab 1: Hello Cloud Run

Duration: 1hour

Objective:

Hello Cloud Run is to provide hands-on experience in deploying a simple containerized application to Google Cloud Run. In this lab, learners will deploy an application using either

the Google Cloud Console or the gcloud CLI and make it accessible via a public HTTPS endpoint. The lab will also introduce key features of Cloud Run, such as automatic autoscaling, traffic management, and version control, as well as how to monitor the service using Cloud Monitoring and Cloud Logging. By the end of the lab, learners will test the deployed application to ensure it works as expected, gaining a solid understanding of how Cloud Run handles deployment and scaling for containerized applications without manual infrastructure management.

Steps:

Task 1: Enable the Cloud Run API and configure your Shell environment

- Enable the Cloud Run API:
- Use Cloud Shell to enable the Cloud Run API.
- If prompted, authorize the use of your credentials.
- You should see a successful message indicating the API was enabled.
- Set the Compute Region:
- Set the region where your Google Cloud resources will be deployed.
- Choose the desired region for your infrastructure (e.g., us-central1).
- Create a LOCATION Environment Variable:
- Create a variable called LOCATION to store the region.
- Assign the name of the region to this variable (e.g., us-central1).

```
Operation "operations/acf.cc11852d-40af-47ad-9d59-477a12847c9e"
finished successfully.
```

Task 2: Write the Sample Application

In this task, you will build a simple Express-based NodeJS application that responds to HTTP requests.

- Create a New Directory:
- In Cloud Shell, create a new directory named helloworld and navigate into it.
- Create and Edit Files:
- You'll create and edit the necessary files using nano or the Cloud Shell Code Editor.
- Create package.json File:
- Create a package.json file and add the necessary configuration, including application details, start script, and Express as a dependency.
- Save package.json File:
- Save the package.json file after adding the required content.
- Create index.js File:
- Create an index.js file where you will configure a basic Express server that responds with "Hello World" when accessed.
- Save index.js File:
- Save the index.js file after adding the server code.

- Note: You can choose to use other programming languages to start with Cloud Run. Find instructions for additional languages in the Quickstarts guide.

Task 3: Containerize Your App and Upload It to Artifact Registry

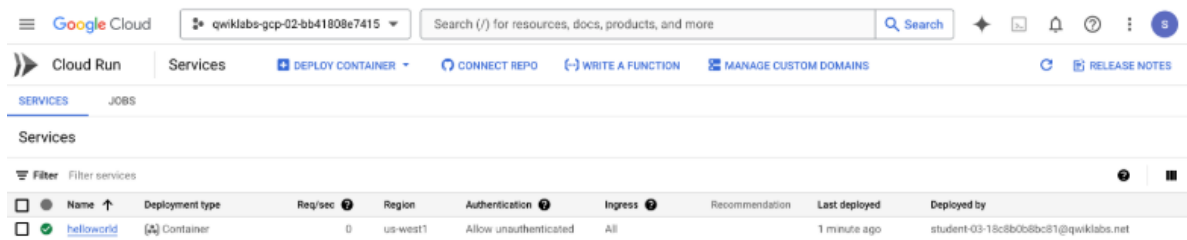
- Create a Dockerfile:
- Create a Dockerfile in the same directory as your source files and define how to build and run the container image.
- Build the Container Image:
- Use Cloud Build to build the container image and tag it with your Google Cloud Project ID.
- Push the Image to Artifact Registry:
- Once the image is built, push it to Artifact Registry for storage and reuse.
- Authenticate Docker with Google Cloud:
- Register gcloud as the credential helper for Docker to allow the push and pull of images from Google Cloud repositories.
- Run the Application Locally:
- Test your containerized app by running it locally in Cloud Shell using Docker commands. You can access it through a web preview or curl.

Task 4: Deploy to Cloud Run

- Deploy to Cloud Run:
- Deploy your containerized application to Cloud Run using the appropriate gcloud command, ensuring it is publicly accessible.
- Confirm the Deployment:
- Once deployed, confirm the service name and wait for the deployment to complete. The service URL will be provided.
- Access the Deployed App:
- Open the service URL in a browser to view your deployed app.

```
Service [helloworld] revision [helloworld-00001-xit] has been
deployed
and is serving 100 percent of traffic.

Service URL: https://helloworld-h6cp412q3a-uc.a.run.app
```



The screenshot shows the Google Cloud console interface. At the top, there's a search bar and navigation links for Cloud Run, Services, and other Google Cloud products. Below the navigation bar, the 'Services' tab is selected, and a table lists the deployed services. The table has columns for Name, Deployment type, Req/sec, Region, Authentication, Ingress, Recommendation, Last deployed, and Deployed by. One service named 'helloworld' is listed with a deployment type of 'Container', 0 Req/sec, in the 'us-west1' region, with 'Allow unauthenticated' authentication and 'All' ingress.

Name	Deployment type	Req/sec	Region	Authentication	Ingress	Recommendation	Last deployed	Deployed by
helloworld	Container	0	us-west1	Allow unauthenticated	All		1 minute ago	student-03-18c8b0b8bc81@qwiklabs.net

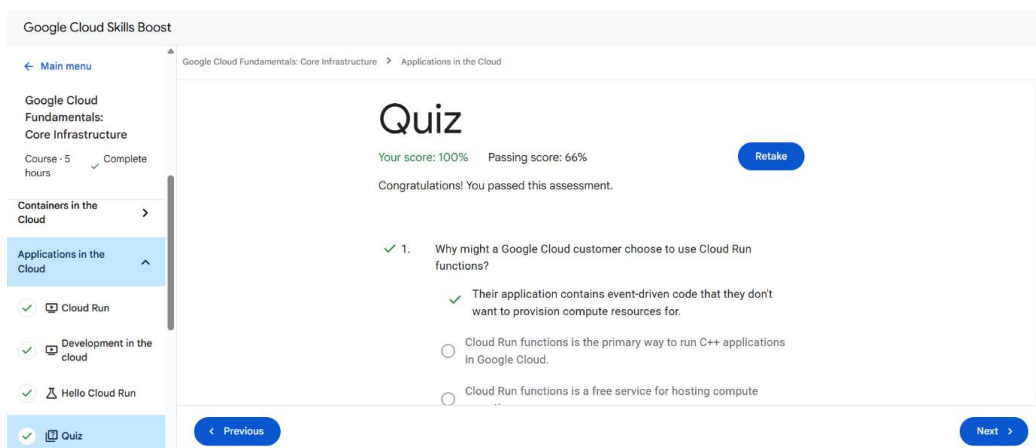
Task 5: Clean Up

- Delete Container Image:
- If desired, delete the helloworld container image from Artifact Registry to avoid unnecessary charges.
- Delete Cloud Run Service:
- Delete the hello world service from Cloud Run.
- End the Lab:
- Once finished, click End Lab to clean up all resources and stop billing for the project.

Congratulations!

You have completed this lab!

Quiz



The screenshot shows the Google Cloud Skills Boost interface. On the left, there's a sidebar with a 'Main menu' and a list of completed courses: 'Google Cloud Fundamentals: Core Infrastructure' (5 hours), 'Containers in the Cloud', 'Applications in the Cloud', 'Cloud Run', 'Development in the cloud', 'Hello Cloud Run', and 'Quiz'. The main content area displays the 'Quiz' results. It shows 'Your score: 100%' and 'Passing score: 66%'. A 'Retake' button is visible. Below this, a message says 'Congratulations! You passed this assessment.' The quiz question is: '1. Why might a Google Cloud customer choose to use Cloud Run functions?'. The correct answer is 'Their application contains event-driven code that they don't want to provision compute resources for.' Other options are 'Cloud Run functions is the primary way to run C++ applications in Google Cloud.' and 'Cloud Run functions is a free service for hosting compute'. Navigation buttons for 'Previous' and 'Next' are at the bottom.

Module 6

Prompt Engineering

Objective:

This module introduces the concept of prompt engineering as a critical technique for maximizing the accuracy, efficiency, and usefulness of generative AI tools. Learners will explore the role of prompt engineering in guiding AI behaviour, reducing ambiguity, and achieving desired outputs across various scenarios such as automating documentation, generating code, analyzing logs, and summarizing large datasets. The module covers core strategies including zero-shot, one-shot, and few-shot prompting, and emphasizes iterative refinement techniques like chain-of-thought prompting. Practical hands-on labs using Google Cloud's Vertex AI platform allow learners to experiment with prompt creation and optimization in real-world contexts. By the end of the module, participants will not only understand how to design and structure prompts effectively but will also gain insight into integrating LLMs into cloud workflows, thereby enhancing decision-making, automation, and productivity in both technical and business environments.

Key Concepts:

- Prompt Engineering:
- The art and science of designing effective inputs (prompts) to guide the output of Large Language Models (LLMs).
- Types of Prompting:
- Zero-shot prompting: Asking the model to complete a task without examples.
- One-shot prompting: Providing one example to guide the model.
- Few-shot prompting: Giving a few examples to improve context and relevance.
- Prompt Structure:
- Importance of clear, specific instructions.
- Role of context, formatting, and tone in output quality.
- Chain-of-Thought Prompting:
- A technique that encourages step-by-step reasoning in the model's response.
- Iterative Prompt Refinement:
- Testing, evaluating, and improving prompts to get better results.
- Generative AI in Google Cloud:
- Usage of Vertex AI and PaLM 2/Gemini models to interact with LLMs.
- Real-world applications like code generation, summarization, data queries, and workflow automation.
- Best Practices:
- Use precise language.
- Avoid ambiguity.
- Guide the model with clear instructions or examples.

My Learning:

- Gained a strong understanding of how Large Language Models respond to different types of prompts.
- Learned to design and structure effective prompts tailored to specific tasks.
- Understood when and how to use zero-shot, one-shot, and few-shot prompting.
- Practiced refining prompts through hands-on labs using Vertex AI Studio.
- Developed skills in applying prompt engineering techniques to real-world cloud scenarios like summarizing logs, generating code snippets, and automating repetitive tasks.
- Acquired insight into how generative AI tools can be used in cloud-based workflows to boost efficiency and productivity.
- Built confidence in experimenting with prompts and optimizing them for better model performance.

Large Language Models

LLMs are **large**, **general-purpose** language models that can be **pre-trained** and then **fine-tuned** for specific purposes.

Large

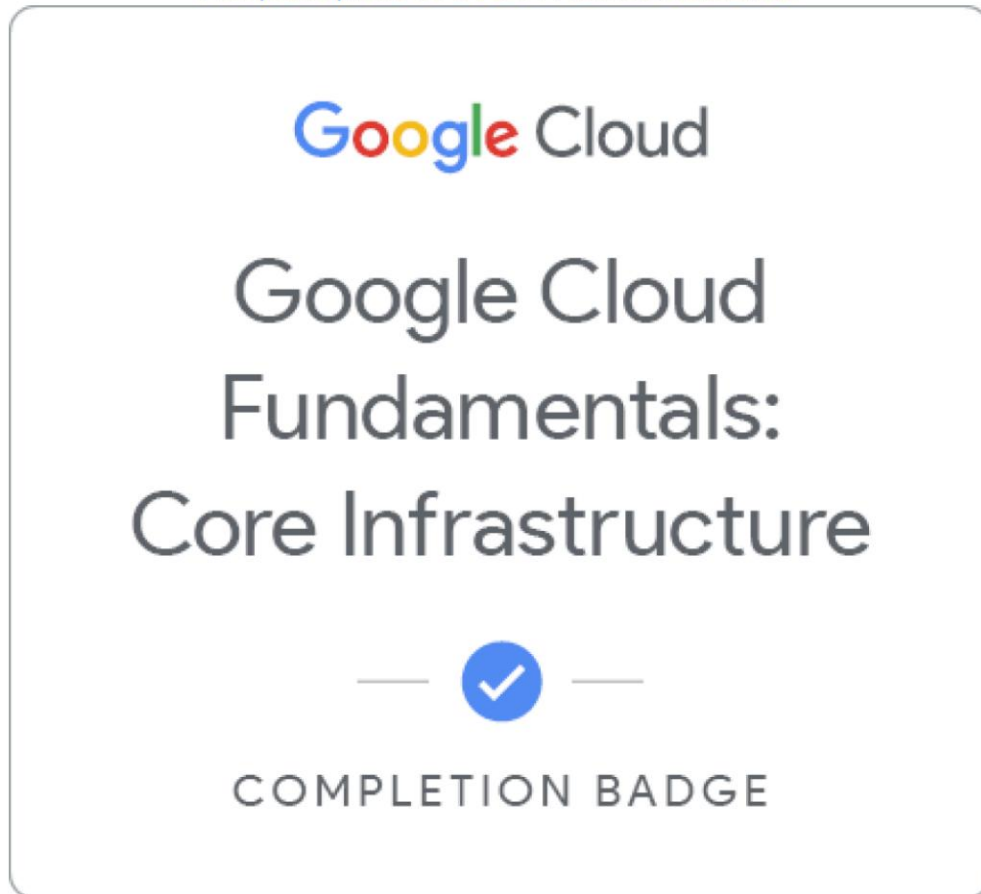
- The size of the training dataset
- The number of parameters



Completion Badge:

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Google Cloud Fundamentals: Core Infrastructure

Mar 31, 2025

Course Platform: Google Cloud Skills Boost

Course3: Get Started with Cloud Storage

Course hours: 2 hours 45 minutes

Course Type: Skill Badge

Learning Objectives

- Understand the fundamentals of Google Cloud Storage
- Create and manage Cloud Storage buckets
- Interact with Cloud Storage using the Command-Line Interface (CLI) and APIs
- Implement Bucket Lock for data protection and compliance
- Demonstrate proficiency in a hands-on, challenge-based assessment environment

Course Overview

This course provided a comprehensive introduction to Google Cloud Storage services, emphasizing both conceptual knowledge and hands-on practice. It was structured around four interactive labs, each focusing on a specific practical aspect of cloud storage management.

- Lab 1 – APIs Explorer: Cloud Storage
Created, managed, and deleted Cloud Storage buckets using the Google APIs Explorer.
- Lab 2 – gsutil Tool
Used the gsutil command-line tool to perform fundamental Cloud Storage operations efficiently.
- Lab 3 – Bucket Lock
Explored Bucket Lock, a feature enabling object retention and ensuring regulatory compliance.
- Lab 4 – Challenge Lab
Applied all previous concepts in a simulated real-world environment, reinforcing understanding through practical assessment.

Lab Completed:

1. APIs Explorer – Cloud Storage

Duration Time: 45 minutes

Overview

The Google APIs Explorer is an interactive tool designed to help users explore and work with Google APIs effectively.

With the APIs Explorer, you can:

- Browse available APIs and versions quickly
 - View available methods and their parameters with inline documentation
 - Execute API requests and view responses in real time
 - Make authenticated and authorized API calls for testing and learning
- Search across all services, methods, and your recent requests to quickly find what you are looking for. Cloud Storage allows worldwide storage and retrieval of any amount of data at any time. You can use Cloud Storage for a range of scenarios including serving website content, storing data for archival and disaster recovery, or distributing large data objects to users via direct download.

In this lab, you will use the APIs Explorer to make Cloud Storage API requests that create and delete Cloud Storage buckets. You will also learn how to copy and delete files in Cloud Storage.

Objective

- Create Cloud Storage buckets using the APIs Explorer
- Upload image files to your Cloud Storage bucket
- Copy an image file and add it to a Cloud Storage bucket using the APIs Explorer
- Delete image files from your Cloud Storage bucket using the APIs Explorer
- Delete a Cloud Storage bucket using the APIs Explorer

Task 1. Create Cloud Storage Buckets

1. To access the Cloud Storage APIs Explorer tool, from the Navigation menu select APIs & Services > Library
2. In the search bar, type Cloud Storage
3. Click on the Google Cloud Storage JSON API from the results list
4. Make sure that API is enabled, if not click Enable
5. Open the Buckets: insert reference. This will open a new tab with the APIs Explorer page loaded

The screenshot displays the 'Request parameters' section of the APIs Explorer. It includes fields for 'project' (filled with 'qwiklabs-gcp-2722fec8f5bbcc77'), 'predefinedAcl', 'predefinedDefaultObjectAcl', 'projection', 'provisionalUserProject' (type 'string'), and 'userProject' (type 'string'). Below these is a 'Show standard parameters' link. The 'Request body' section shows a JSON object with a single key 'name' and value 'qwiklabs-bucket01'. A note at the bottom states: 'For suggestions, press control+space or click one of the blue "add" circles.'

```
Request parameters
project
qwiklabs-gcp-2722fec8f5bbcc77
predefinedAcl
predefinedDefaultObjectAcl
projection
provisionalUserProject
string
userProject
string
Show standard parameters
Request body
{
  "name": "qwiklabs-bucket01"
}
```

For suggestions, press control+space or click one of the blue "add" circles.

7. Click the Execute button.
8. Select the student account you started the lab with.
9. On the next screen, click Allow to give APIs Explorer access.

```
{
  "kind": "storage#bucket",
  "id": "qwiklabs-bucket01",
  "selfLink": "https://www.googleapis.com/storage/v1/b/qwiklabs-bucket01",
  "projectNumber": "250399850182",
  "name": "qwiklabs-bucket01",
  "timeCreated": "2019-10-18T13:59:08.300Z",
  "updated": "2019-10-18T13:59:08.300Z",
  "metageneration": "1",
  "iamConfiguration": {
    "bucketPolicyOnly": {
      "enabled": false
    },
    "uniformBucketLevelAccess": {
      "enabled": false
    }
  },
  "location": "US",
  "locationType": "multi-region",
  "storageClass": "STANDARD",
  "etag": "CAE="
}
```

Task 2. Make a second Cloud Storage bucket

Now make another Cloud Storage bucket to get hands-on practice copying files between the two.

1. Still, in the insert method, ensure that your Project ID is still in the project field.
2. In the request body, for the name key-value pair, give your second bucket a unique name.
3. Make sure that there are no trailing spaces in any of the fields.

4. Click the Execute button. Your response should resemble the following:

```
{
  "kind": "storage#bucket",
  "id": "qwiklabs-bucket02",
  "selfLink": "https://www.googleapis.com/storage/v1/b/qwiklabs-
bucket02",
  "projectNumber": "250399850182",
  "name": "qwiklabs-bucket02",
  "timeCreated": "2019-10-18T13:59:08.300Z",
  "updated": "2019-10-18T13:59:08.300Z",
  "metageneration": "1",
  "iamConfiguration": {
    "bucketPolicyOnly": {
      "enabled": false
    },
    "uniformBucketLevelAccess": {
      "enabled": false
    }
  },
  "location": "US",
  "locationType": "multi-region",
  "storageClass": "STANDARD",
  "etag": "CAE="
}
```

5. Cloud Storage: Qwik Start - CLI/SDK – Used **gsutil** to perform storage operations from the command line.
6. Google Cloud Storage - Bucket Lock – Implemented and configured Bucket Lock for compliance and data protection.
7. Get Started with Cloud Storage: Challenge Lab – Applied acquired skills in a timed, assessment-based lab to demonstrate proficiency.

Task 3: Upload Files to Your Cloud Storage Bucket

Objective:

To practice uploading files to a Cloud Storage bucket using the Cloud Storage browser in the Google Cloud Console.

Steps to Complete the Task:

1. Download Image Files:

- Save the following two images to your computer:
 - **demo-image1.png**: A picture of a dog.
 - **demo-image2.png**: A public domain image of Ada Lovelace.

2. Open Cloud Storage Browser:

- Navigate to the Cloud Storage section in the Google Cloud Console.
- Select the first bucket from the list (e.g., **qwiklabs-bucket01**).

3. Upload Files:

- Click on the **Upload files** button.
- Select and upload both **demo-image1.png** and **demo-image2.png** from your computer.

4. Verify Upload:

- Go to the **Objects** tab of the selected bucket.
- Ensure both files (**demo-image1.png** and **demo-image2.png**) are listed.

Outcome:

The bucket should now contain both image files. The Objects tab of the selected Cloud Storage bucket will display:

- demo-image1.png
- demo-image2.png

Task 4: Copy Files Between Cloud Storage Buckets

Objective:

To learn how to copy an object from one Cloud Storage bucket to another using the

Objects: copy method available in the Google Cloud API Explorer.

Steps to Complete the Task:

1. Navigate to the API Reference:

- Search for and use the **Objects: copy** reference in the **API Explorer**.

2. Fill in Request Parameters:

- **sourceBucket**: Enter the name of the bucket containing **demo-image1.png**.
- **sourceObject**: Enter **demo-image1.png**.
- **destinationBucket**: Enter the name of your second (empty) bucket.

- **destinationObject:** Enter `demo-image1-copy.png`.

3. Credentials Configuration:

- Under the **Credentials** section, ensure the following are selected:
 - Google OAuth 2.0
 - API Key
- Click the question mark icon next to the **Credentials** title for additional FAQs, if needed.
- Ensure there are **no trailing spaces** in any of the fields.

4. Execute the Method:

- Scroll down and click **Execute** to run the copy operation.

Expected Output:

After executing the request, a JSON response will confirm that the file has been successfully copied. A sample response resembles the following:

```
{
  "kind": "storage#object",
  "id": "qwiklabs-bucket02/demo-image1-copy.png/1571408245199237",
  "selfLink": "https://www.googleapis.com/storage/v1/b/qwiklabs-bucket02/o/demo-image1-copy.png",
  "name": "demo-image1-copy.png",
  "bucket": "qwiklabs-bucket02",
  "generation": "1571408245199237",
  "metageneration": "1",
  "contentType": "image/png",
  "timeCreated": "2019-10-18T14:17:25.198Z",
  "updated": "2019-10-18T14:17:25.198Z",
  "storageClass": "STANDARD",
  "timeStorageClassUpdated": "2019-10-18T14:17:25.198Z",
  "size": "401951",
  "md5Hash": "LbpHpwhnApQKQx9IEXjTsQ==",
  "mediaLink": "https://www.googleapis.com/download/storage/v1/b/qwiklabs-bucket02/o/demo-image1-copy.png?generation=1571408245199237&alt=media",
  "owner": {
    "entity": "user-gcpstaging93416_student@qwiklabs.net"
  },
  "crc32c": "j5oPrg=",
  "etag": "CIWjgvL/peUCEAE="
}
```

Task 5: Delete Files from a Cloud Storage Bucket

Objective:

Delete objects from a Cloud Storage bucket using the

Objects: delete method in the API Explorer.

Steps:

1. Open API Explorer

Navigate to:

APIs & Reference > JSON API > Objects > delete

2. Delete **demo-image1.png**

- **bucket:** Enter the name of your bucket
- **object:** **demo-image1.png**
- Ensure **Google OAuth 2.0** and **API Key** are selected
- Remove any trailing spaces and click **Execute**
- A **204** status code confirms the deletion

3. Delete **demo-image2.png**

- Change **object** to **demo-image2.png** (bucket remains the same)
- Click **Execute**
- Again, a **204** status code confirms success

4. Verify in Console

- Go to **Buckets** in the Cloud Console
- Select your bucket to confirm both files are deleted

Outcome:

Both image files were successfully removed from the bucket using the API.

Task 6: Delete Your Cloud Storage Bucket

Objective:

Delete an empty Cloud Storage bucket using the

Buckets: delete method in the API Explorer.

Steps:

1. Open API Explorer

Navigate to:

APIs & Reference > JSON API > Buckets > delete

2. Delete the Bucket

- **bucket:** Enter the name of your first (empty) bucket
- Ensure **Google OAuth 2.0** and **API Key** are selected
- Remove any trailing spaces and click **Execute**
- A **204** status code confirms the bucket was deleted

3. Verify in Console

- Return to the **Cloud Console**
- Go to **Buckets**
- Confirm the first bucket is no longer listed

Outcome:

The Cloud Storage bucket was successfully deleted using the `buckets.delete` method.

2. Cloud Storage – Qwik Start (CLI/SDK)

Duration Time: 30 minutes

Overview:

This hands-on lab introduced the core functionalities of **Google Cloud Storage** using the **Google Cloud command-line interface (CLI)**. It covered essential operations such as creating buckets, managing folder structures, uploading files, and setting access permissions—all performed via command-line tools. Cloud Storage supports worldwide storage and retrieval of large volumes of data. It's useful for a variety of use cases including hosting static content, backup and archival, and distributing large data files.

Key Concepts:

- Creating a Cloud Storage bucket via CLI
- Uploading objects to the bucket
- Creating folders and subfolders within the bucket
- Making stored objects publicly accessible

Outcome:

By the end of the lab, all key actions were successfully executed and verified in the Cloud Console, demonstrating proficiency in managing Cloud Storage using the CLI/SDK environment.

Task 1: Create a Bucket

Objective:

Create a Cloud Storage bucket using the Google Cloud CLI (`gcloud`) while following the global bucket naming rules.

Steps:

1. Review Bucket Naming Rules:

- It must be globally unique and publicly visible.
- Use only lowercase letters, numbers, dashes (-), underscores (_), and dots (.).
- It must start and end with a letter or number.
- Length: 3–63 characters (up to 222 if using dots, with each part ≤ 63).

- Must not resemble IP addresses or use "goog"/"google".

Create the Bucket: Run the following command with a unique bucket name:
gcloud storage buckets create gs://<YOUR-BUCKET-NAME>

2. Verify Creation:

- If successful, the bucket will appear in **Cloud Console > Cloud Storage > Buckets**.
- To view default settings, click the bucket name and open the **Configuration** tab.

3. Error Handling:

- If the bucket name is already taken, a **409 Bucket already exists** error is returned.
- Retry with a different name.

Outcome:

A new Cloud Storage bucket was successfully created using the **gcloud** CLI, demonstrating understanding of naming rules and command-line operations.

Task 2: Upload an Object into Your Bucket

Objective:

Upload an object (image file) to a Cloud Storage bucket using the command-line interface in Cloud Shell.

Steps:

Download the Image: Use **curl** to download the image locally:

```
curl
https://upload.wikimedia.org/wikipedia/commons/thumb/a/a4/Ada_Love1
Ada_Lovelace_portrait.jpg --output ada.jpg
```



Upload to Cloud Storage: Transfer the image to your bucket:

```
gcloud storage cp ada.jpg gs://YOUR-BUCKET-NAME
```



Clean Up: Delete the local copy to keep the environment tidy:
rm ada.jpg

Outcome:

The image **ada.jpg** was successfully uploaded to the Cloud Storage bucket via the command line.

Task 3: Download an Object from Your Bucket

Objective:

Retrieve an object from the Cloud Storage bucket using the `gcloud` CLI.

Steps:

Download

the

Image:

```
gcloud storage cp -r gs://YOUR-BUCKET-NAME/ada.jpg .
```



Result:

- The image is successfully copied from the bucket to Cloud Shell.
- A confirmation message confirms the completion of the operation.

Outcome:

The image was downloaded from the Cloud Storage bucket to the local Cloud Shell environment, verifying the upload in the previous task.

Task 4: Copy an Object to a Folder in the Bucket

Objective:

Copy an object (image file) to a new folder inside the Cloud Storage bucket using the `gcloud` CLI.

Steps:

1. **Create a Folder and Copy the Image:** Use the following command to create a folder (`image-folder`) and copy the image (`ada.jpg`) into it:

```
gcloud storage cp gs://YOUR-BUCKET-NAME/ada.jpg gs://YOUR-BUCKET-NAME/image-folder/
```



Outcome:

The image file `ada.jpg` was successfully copied into a new folder (`image-folder`) inside the Cloud Storage bucket

Task 5: List Contents of a Bucket or Folder

Objective:

List the contents of a Cloud Storage bucket (or folder) using the **gcloud** CLI.

Steps:

1. **List Bucket Contents:** Use the following command to list all objects in your bucket:

```
gcloud storage ls gs://YOUR-BUCKET-NAME
```



Outcome:

The contents of the Cloud Storage bucket were successfully listed, showing both the image file and the folder.

Task 6: List Details for an Object

Objective:

Retrieve detailed information about an object in a Cloud Storage bucket using the **gcloud** CLI.

Steps:

1. **List Object Details:** Use the following command to view details such as size and creation date for the uploaded image:

```
gcloud storage ls -l gs://YOUR-BUCKET-NAME/ada.jpg
```



Outcome:

The details for the **ada.jpg** image were successfully retrieved, showing its size and creation date.

Task 7: Make Your Object Publicly Accessible

Objective:

Grant all users read permission to an object in your Cloud Storage bucket, making it publicly accessible.

Steps:

1. **Grant Read Permission:** Use the **gsutil acl ch** command to set the object's Access Control List (ACL) to allow all users read access:

```
gsutil acl ch -u AllUsers:R gs://YOUR-BUCKET-NAME/ada.jpg
```



Outcome:

The `ada.jpg` image is now publicly accessible, allowing anyone with the link to view the file.

Task 8: Remove Public Access

Objective:

Remove public read access from an object in your Cloud Storage bucket.

Steps

1. **Remove Public Access:** Use the `gsutil acl ch` command to revoke the public read access:

```
gsutil acl ch -d AllUsers gs://YOUR-BUCKET-NAME/ada.jpg
```



2.Result: The command will confirm the removal of public access:

```
Updated ACL on gs://YOUR-BUCKET-NAME/ada.jpg
```

3.Verification: Refresh the console to ensure that the checkmark for public access is removed.

Objective:

Delete an object from your Cloud Storage bucket.

Steps

1. **Delete the Object:** Use the `gcloud storage rm` command to remove the `ada.jpg` file:

```
gcloud storage rm gs://YOUR-BUCKET-NAME/ada.jpg
```



2. Verification: Refresh the console, and the image file will no longer be listed in the bucket, though any copy in folders (e.g., `image-folder/`) may remain.

3. Google Cloud Storage - Bucket Lock Lab

Duration Time: 40 minutes

Overview:

In this lab, you'll learn how to use the **Cloud Storage Bucket Lock** feature, which allows you to enforce a **data retention policy** for a Cloud Storage bucket. This policy governs how long objects must be retained before they can be modified or deleted. Once locked, the retention policy cannot be reduced or removed.

The **Bucket Lock** feature is especially important for **regulatory compliance**, ensuring that data is kept for a specified period in industries such as finance and healthcare. Combined with detailed audit logging and Object Lifecycle Management, Bucket Lock helps meet industry standards such as **FINRA**, **SEC**, **CFTC**, and healthcare data retention regulations.

Key Concepts:

- **Data Retention Policy:** Specifies the duration for which objects must be retained.
- **Bucket Lock:** Locks the retention policy, preventing any changes to it.
- **Regulatory Compliance:** Ensures data retention meets industry regulations.

By the end of this lab, you will know how to:

1. **Create a Cloud Storage Bucket**
2. **Define and Implement an Object Retention Policy**
3. **Lock and Remove a Retention Policy**

Task 1: Create a New Bucket

1. **Define the Environment Variable:** Use your **Google Cloud Project ID** as the bucket name:

```
export BUCKET=$(gcloud config get-value project)
```



2. **Create the Bucket:** Use the **gsutil** command to create a new Cloud Storage bucket:

```
gsutil mb "gs://$BUCKET"
```



Task 2: Define a Retention Policy

1. **Set the Retention Policy:** Apply a retention policy of 10 seconds to the bucket:

```
gsutil retention set 10s "gs://$BUCKET"
```



2. Verify the Retention Policy: Check the retention policy settings for the bucket:

```
gsutil retention get "gs://$BUCKET"
```



3. Upload a Sample Object: Add a dummy transaction record:

```
gsutil cp gs://splis/gsp297/dummy_transactions "gs://$BUCKET/"
```



4. Review Retention Expiration: Get details of the object and check when it expires:

```
gsutil ls -L "gs://$BUCKET/dummy_transactions"
```



Task 3: Lock the Retention Policy

1. Lock the Retention Policy: Prevent any changes to the retention policy by locking it:

- **gsutil retention lock "gs://\$BUCKET/"**

2. Confirm the Lock: You'll be prompted to confirm the lock

3. Verify the Locked Policy: Confirm the retention policy is locked:

- **gsutil retention get "gs://\$BUCKET"**

Task 4: Temporary Hold

1. Set a Temporary Hold: Prevent the object from being deleted during an audit:

- **gsutil retention temp set "gs://\$BUCKET/dummy_transactions"**

2. Attempt to Delete the Object: This should fail if the hold is active:

- **gsutil rm "gs://\$BUCKET/dummy_transactions"**

3. Release the Temporary Hold: Once the audit is complete:

- `gsutil retention temp release "gs://$BUCKET/dummy_transactions"`

4. Delete the Object: After the retention policy expires:

- `gsutil rm "gs://$BUCKET/dummy_transactions"`

Task 5: Event-Based Holds

1. Enable Default Event-Based Hold:

- `gsutil retention event-default set "gs://$BUCKET/"`

2. Upload a Loan Record:

- `gsutil cp gs://spl/gsp297/dummy_loan "gs://$BUCKET/"`

3. Verify Event-Based Hold:

- `gsutil ls -L "gs://$BUCKET/dummy_loan"`

4. Release the Event-Based Hold (e.g., loan paid off):

- `gsutil retention event release "gs://$BUCKET/dummy_loan"`

5. Verify Retention Expiration:

- `gsutil ls -L "gs://$BUCKET/dummy_loan"`

Task 6: Remove a Retention Policy

1. Delete the Empty Bucket:

- `gsutil rb "gs://$BUCKET/"`

Unfortunately, the branch shuts down its lending operations. The Branch IT Administrator still needs to maintain the existing records for their full duration, but no longer expects to produce records in the bucket. After the last loan Retention Period has expired and no longer subject to a hold, the Branch IT Administrator can then delete the empty bucket. The bucket can be deleted even though it has a locked Retention Policy because it contains no data subject to retention.

4. Report: Get Started with Cloud Storage - Challenge Lab

Duration Time: 45 minutes

Overview

This report outlines my experience completing the "Get Started with Cloud Storage: Challenge Lab", part of the Google Cloud training series. The lab is designed to assess practical skills in Google Cloud Storage, specifically without guided, step-by-step instructions. Instead, I was given a scenario and a set of tasks to complete within a 45-minute window.

This lab functioned as a challenge-based assessment, where success depended on applying previously learned concepts from earlier labs to complete the tasks independently. The challenge tested my understanding of Cloud Storage services, including bucket and object management, access configuration, and policy enforcement.

Objective

The primary objective of the challenge lab was to complete all provided tasks correctly within the given time frame. The tasks simulated real-world cloud storage scenarios and required knowledge of Cloud Storage tools and administrative commands.

To succeed, I had to demonstrate the ability to:

- Deploy and manage Google Cloud Storage resources
- Configure permissions and access controls
- Use the command-line interface efficiently (`gcloud`, `gsutil`)
- Apply retention and lifecycle policies to stored data

Skills Demonstrated

The challenge allowed me to put into practice a range of Google Cloud Storage skills, including:

- Creating buckets using the Cloud SDK
- Uploading files and copying them between folders
- Modifying and revoking object access (public and private)
- Viewing metadata and configuration details of stored objects
- Setting up and locking object retention policies

These tasks required efficient use of the following command-line tools:

- `gcloud storage`
- `gsutil`
- Linux environment variables and command scripting

Problem-Solving Approach

Since this was a challenge lab, I was required to:

- Analyze the given scenario and infer appropriate solutions
- Use documentation when necessary to troubleshoot command errors
- Apply logical reasoning to modify object states and bucket properties
- Cross-check results using `ls`, `cp`, `rm`, and `retention` flags in `gsutil`

I had to pay close attention to command syntax and output messages to determine whether operations were successful or needed correction.

Evaluation and Completion

The lab featured an automated scoring system that provided real-time progress tracking. I was able to monitor which tasks were complete and which still required attention. This system helped me validate that all tasks were completed correctly before the deadline.

To receive 100% completion:

- I ensured that all tasks were executed successfully.
- I confirmed all permissions, policies, and object states matched the lab requirements.
- I completed the lab within the 45-minute limit.

Course Summary

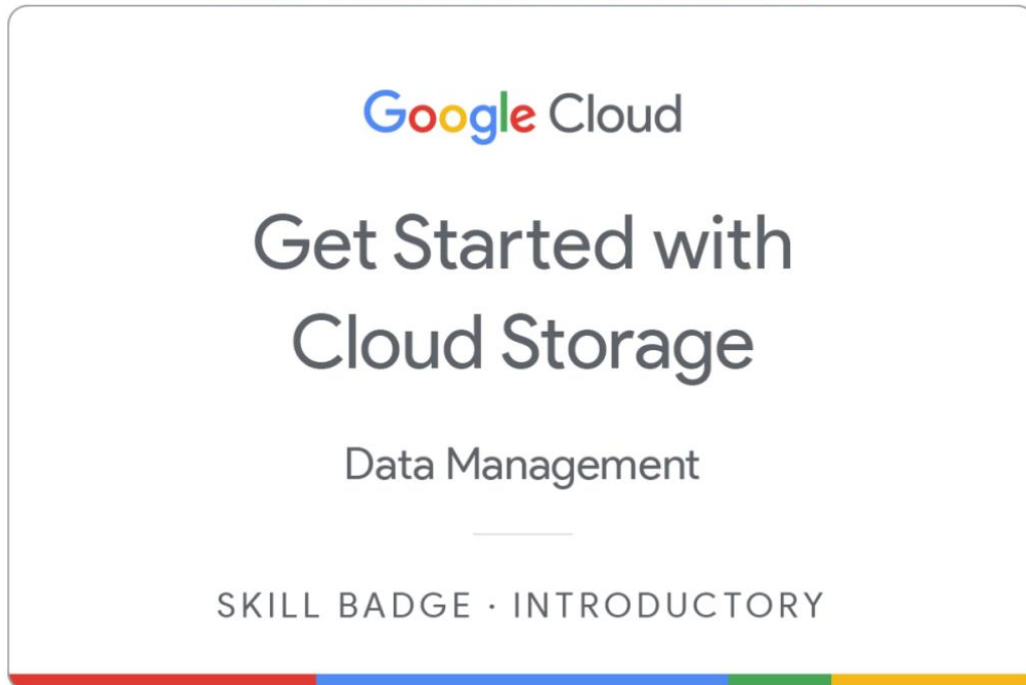
Completing the "Get Started with Cloud Storage: Challenge Lab" was a valuable exercise in consolidating my understanding of Cloud Storage fundamentals. It tested my ability to operate independently in a cloud environment, apply theoretical knowledge practically, and troubleshoot technical issues without reliance on instructions.

This lab served as a strong capstone to the foundational coursework and provided confidence in handling real-world cloud storage operations on the Google Cloud Platform.

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Get Started with Cloud Storage

Apr 1, 2025